

Low-rank quaternion tensor completion by truncated nuclear norm regularization for color video recovery

Han Jiao, Qilong Liu*, Kun Liang, Liyuan An

School of Mathematical Sciences, Guizhou normal university, 550025, PR China

ARTICLE INFO

Communicated by L. Shen

Keywords:

Quaternion
Color videos
Truncated nuclear norm
Tensor completion
Low-rank

ABSTRACT

Low-rank quaternion tensor completion has attracted significant attention for recovering incomplete visual data with missing elements. While the singular value decomposition (SVD) of quaternion tensors effectively characterizes the tensor low-rankness, existing approaches relying on transform-based tensor products substantially increase the computational complexity in the completion procedure. To address this limitation, we directly employ the t-product to examine the tensor completion. The proposed model establishes a truncated nuclear norm through quaternion tensor SVD to effectively capture the low-rank structural features. The fast quaternion tensor singular value decomposition (FQT-SVD) based on t-product and the alternating direction method of multipliers (ADMM) are used to solve the model. Numerical experiments on color video recovery demonstrate our method's superior performance and efficiency compared to state-of-the-art approaches.

1. Introduction

With the continuous advancement of computer vision processing, recovering missing color videos has become a critical area of research. Due to various unpredictable or unavoidable factors caused by the acquisition and transmission processes, only partial observations of video data are obtained in many real-world applications. The main aim of color video restoration is to retrieve missing components from incomplete color videos based on limited observations. Color videos, being a form of high-dimensional data, can be effectively modeled as tensors, which extend the concepts of matrices and vectors. Using the tensor representation, color video restoration can be reformulated as a tensor completion (TC) problem. The core of the TC problem lies in how to build a proper low-rank regularization to accurately measure the global structure of the underlying data [1,2]. The low-rank tensor completion problem has been extensively researched and demonstrated significant utility in the applications of image and video processing [3,4], machine learning [5,6], data mining [7,8], and computer vision [9,10].

In the context of color image and video restoration, a color image contains red (R), green (G), and blue (B) channels. Traditional matrix-based or tensor-based completion methods usually ignore the interconnection between channels, as these recovery methods are applied independently to the red, green, and blue channels, potentially

resulting in color distortion during the recovery process [11]. To overcome this drawback, the quaternion representation of the RGB image was proposed by Pei [12] and Sangwine [13]. The quaternion matrix and tensor were used to represent color images and videos [14–16]. In more detail, the pixels in each frame are encoded on three imaginary parts of a quaternion as an element in a quaternion matrix, and each quaternion matrix is then stacked into a quaternion tensor. This means that the inherent color structures of color images and color videos can be completely preserved [17,18]. The visual representation of a three-dimensional quaternion tensor that represents a color video is shown in Fig. 1. The quaternion has achieved excellent results in color image processing problems, including histopathological image analysis [19], color image denoising and representation [20,21], color object recognition [22,23], and color image or video recovery problems [24,25].

Existing works have proven that exact TC greatly benefits from the low-rankness of a tensor [16,25,26]. Since the definition of tensor rank is not unique, each of which has its own limitations, the tensor rank is not well-defined. For instance, the CANDECOMP/PARAFAC (CP) rank, is defined as the minimum number of rank-1 tensors used to express a tensor. Although the CP rank is consistent with the definition of rank of matrix, determining the convex relaxation of the CP rank is also quite challenging when solving practical problems, which makes CP-rank-based tensor recovery difficult. Meanwhile, it is generally NP-hard to compute

* Corresponding author.

Email addresses: 17385190210@163.com (H. Jiao), qlliu@gznu.edu.cn (Q. Liu), kliang8629@163.com (K. Liang), anly0811@163.com (L. An).

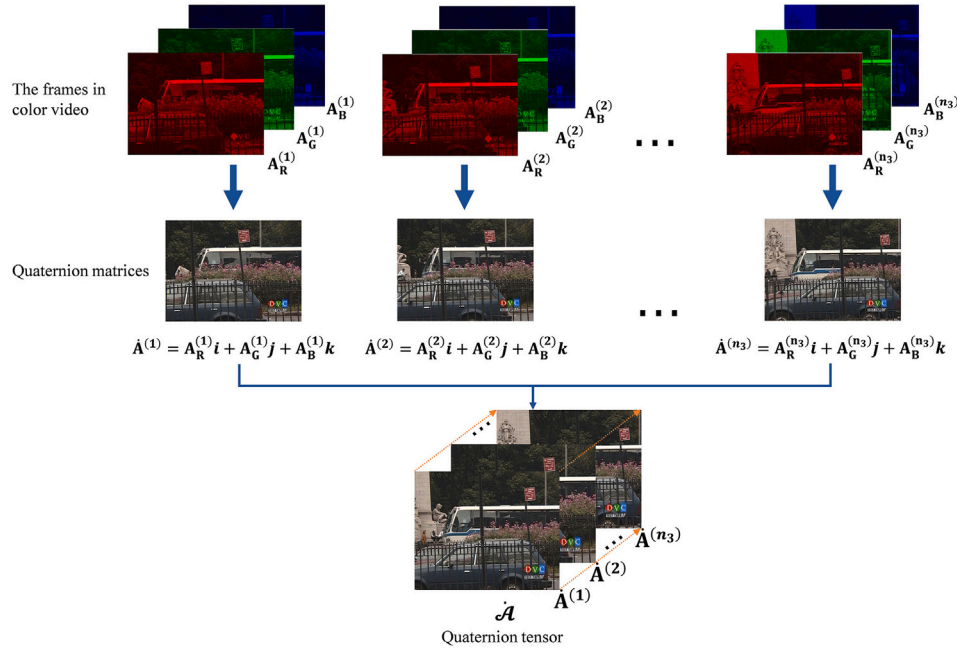


Fig. 1. The color video is depicted using a three-dimensional quaternion tensor.

the CP rank of a tensor data accurately [27,28]. Another tensor decomposition rank, the Tucker rank, is defined as a multilinear rank constitutes of ranks of all matrices unfolded along each mode of a tensor, which is induced by the Tucker decomposition [29,30]. It can be computed efficiently by matrix singular value decomposition. However, the Tucker rank is a multi-variable that markedly differs from matrix rank and has practical application challenges. Furthermore, during the process of unfolding a tensor to a matrix along one mode, the structure of high-dimensional data is inevitably destroyed. Besides, the Sum-of-Nuclear-Norms (SNN) [31], commonly used as a convex surrogate for the Tucker rank, is not the tightest convex relaxation.

Moreover, a novel tensor decomposition scheme, the tensor singular value decomposition t-SVD induced by tensor-tensor product (t-product) was proposed in [32]. The t-product operator was initially motivated by desiring a closed operation that preserves the order of a tensor. It is a new notion of multiplication between tensors. The t-SVD factorizes a tensor into the t-product of two orthogonal tensors and one singular value tensor. Based on the t-SVD, a new tensor rank, *i.e.*, the tensor tubal rank, is introduced [33], which is defined as the number of nonzero singular tubes of the singular value tensor. In comparison to the previously mentioned tensor decomposition strategies, the novel t-SVD scheme has proven superior in capturing the global structural information prevalent in various tensor datasets [34,35]. The tensor tubal rank can effectively characterize the intrinsic low-rank structure of a tensor, and the tensor model derived from the tubal rank offers greater and more flexible modeling capabilities compared to those based on other tensor ranks, especially for those tensors that have fixed orientation or certain spatial shifting. Therefore, it is desirable to apply the t-SVD to color video restoration.

Based on the t-SVD scheme, Zhang et al. [34] introduced the tubal nuclear norm (Tubal-NN), which is the sum of the nuclear norms of all frontal slices in the Fourier domain. Tubal-NN serves as a convex relaxation of the ℓ_1 -norm associated with tensor rank. Nevertheless, Hu et al. [36] clarified that the nuclear norm may not effectively approximate the rank function, since it simultaneously minimizes all of the singular values. Note that the larger the singular value, the less it will affect the rank. They further proposed the truncated nuclear norm regularization (TNNR) as a more accurate and much tighter substitute to

the rank function. By adopting the TNNR method, Liu et al. [37] devised a weighted TNNR using a gradient descent scheme. Lee and Lam [38] exploited the ghost-free high dynamic range imaging from irradiance maps. Additionally, Xue et al. [39] proposed the tensor truncated nuclear norm (T-TNN) for a superior estimation of the rank.

In the quaternion field, low-rank quaternion tensor completion has emerged as a powerful approach for color image and video restoration [16,25,40,41], achieving remarkable performance. However, the major limitation of the existing approaches based on other decomposition strategies is that the rank may not be well approximated in practice. For instance, under the definition of Tucker rank, Miao et al. [16] proposed a low-rank quaternion tensor completion method, which achieved a better performance compared to traditional tensor-based approaches. Nevertheless, during the process of unfolding a tensor to a matrix along one mode, the structure of high-dimensional data is inevitably destroyed. Besides, the upper bound for each individual rank is usually small and may not be suitable for describing the global information of a tensor [42]. In [41], the quaternion tensor train decomposition was first proposed, but the rank may not be well approximated by it similar to the tubal rank in practice. To greatly preserve the color information and inherent spatial structure of the video, it is expected to extend t-SVD to the quaternion field. Yang et al. [25] achieved the low-rank quaternion tensor completion using the novel transform-based tensor product and corresponding transform-based quaternion t-SVD, but it greatly increases the complexity of the completion problem. Recently, Pan et al. [43] proved that a quaternion circulant matrix can be block-diagonalized into 1×1 and 2×2 blocks via the permuted discrete quaternion Fourier transform matrix. They utilized this block-diagonalized form to study fast quaternion tensor singular value decomposition (FQT-SVD). This means that we no longer need to find the transform-based tensor product and corresponding transform-based quaternion t-SVD for quaternion tensor completion. Furthermore, we directly employ the t-product to examine the tensor completion. We use the truncated nuclear norm to characterize the low-rankness of the quaternion tensor.

Motivated by the above concerns, based on FQT-SVD and the TNNR method, we provide the definition of Quaternion Tensor Truncated Nuclear Norm (QT-TNN), and then propose a novel low-rank quaternion

tensor completion method for color video recovery in this paper. It may produce better recovery results compared to the typical approaches induced by other types of quaternion tensor rank minimization for video recovery.

The main contributions of this work are summarized:

- This work focuses on quaternion tensor recovery and develops a new quaternion tensor completion model called quaternion tensor truncated nuclear norm regularization (QT-TNNR), which depicts the low-rankness of the target quaternion tensor. By processing color visual data in the quaternion field, it prevents corruption of the data structure.
- By introducing a nuclear norm truncation strategy, the low-rankness is depicted as a more precise substitute for the rank function, which is based on FQT-SVD and TNNR.
- This model exploits FQT-SVD and the alternating direction method of multipliers (ADMM) [44] in optimization. By alternately taking the two steps above, the experiments conducted on color video recovery indicate that the proposed method outperforms both state-of-the-art and comparable tensor-based and quaternion tensor-based approaches.

The remainder of this paper is organized as follows. Section 2 presents essential notations and preliminaries. Section 3 proposes the QT-TNNR model. In Section 4, comprehensive experiments on five standard color video databases are presented to demonstrate the feasibility and effectiveness of the proposed algorithm. The conclusions are provided in Section 5.

2. Notations and preliminaries

In this section, we first review quaternion algebra. Next, we introduce essential notations and basic preliminaries that will be used in the sequel.

2.1. Quaternion algebra

In this paper, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ denote the real number field, the complex number field, and the quaternion field, respectively. The quaternion is typically expressed in the following Cartesian form,

$$\dot{q} = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k},$$

where $q_0, q_1, q_2, q_3 \in \mathbb{R}$, and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are imaginary units such that

$$\begin{cases} \mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{i}\mathbf{j}\mathbf{k} = -1, \\ \mathbf{i}\mathbf{j} = -\mathbf{j}\mathbf{i} = \mathbf{k}, \mathbf{j}\mathbf{k} = -\mathbf{k}\mathbf{j} = \mathbf{i}, \mathbf{k}\mathbf{i} = -\mathbf{i}\mathbf{k} = \mathbf{j}. \end{cases} \quad (1)$$

The quaternion $\dot{q}$ can be simply written as $\dot{q} = \text{Re } \dot{q} + \text{Im } \dot{q}$ with real component $\text{Re } \dot{q} = q_0$ and imaginary component $\text{Im } \dot{q} = q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$. When the real component $\text{Re } \dot{q} = 0$, we call a quaternion $\dot{q}$ pure quaternion. Given two quaternions $\dot{p}$ and $\dot{q} \in \mathbb{H}$, the sum and multiplication of them are

$$\dot{p} + \dot{q} = (p_0 + q_0) + (p_1 + q_1)\mathbf{i} + (p_2 + q_2)\mathbf{j} + (p_3 + q_3)\mathbf{k}$$

and

$$\begin{aligned} \dot{p}\dot{q} = & (p_0q_0 - p_1q_1 - p_2q_2 - p_3q_3) + (p_0q_1 + p_1q_0 + p_2q_3 - p_3q_2)\mathbf{i} \\ & + (p_0q_2 - p_1q_3 + p_2q_0 + p_3q_1)\mathbf{j} + (p_0q_3 + p_1q_2 - p_2q_1 + p_3q_0)\mathbf{k}. \end{aligned}$$

The definitions of the quaternion conjugate $\dot{q}^*$ and the modulus $|\dot{q}|$ of $\dot{q}$ are denoted as follows

$$\dot{q}^* = \text{Re } \dot{q} - \text{Im } \dot{q}, \quad |\dot{q}| = \sqrt{\dot{q}\dot{q}^*} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2}.$$

We call a quaternion is a unit quaternion if its modulus equals 1, i.e., $|\dot{q}| = 1$. Similarly, for quaternion matrix $\dot{\mathbf{Q}} = (\dot{q}_{uv}) \in \mathbb{H}^{m \times n}$, its transpose and conjugate-transpose are denoted as $\dot{\mathbf{Q}}^T = (\dot{q}_{vu}) \in \mathbb{H}^{n \times m}$ and $\dot{\mathbf{Q}}^H = (\dot{q}_{vu}^*) \in \mathbb{H}^{n \times m}$, respectively.

2.2. Notations

In this paper, a scalar, a vector, a matrix, and a tensor are denoted as a , $\mathbf{a}$, $\mathbf{A}$, and $\mathcal{A}$, respectively. A dot (above the variable) is used to denote a quaternion variable, $\dot{a}$, $\dot{\mathbf{a}}$, $\dot{\mathbf{A}}$, and $\dot{\mathcal{A}}$ represent a quaternion scalar, a quaternion vector, a quaternion matrix, and a quaternion tensor, respectively. For a three-dimensional tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$, its (i, j, k) th element is denoted as $\mathcal{A}_{ijk}$. The i th horizontal, lateral, and frontal slices of $\mathcal{A}$ are denoted by the MATLAB notations $\mathcal{A}(i, :, :)$, $\mathcal{A}(:, i, :)$, and $\mathcal{A}(:, :, i)$, respectively. Specifically, the frontal slice $\mathcal{A}(:, :, i)$ is briefly denoted as $\mathbf{A}^{(i)}$. The inner product of matrices $\mathbf{A}$ and $\mathbf{B}$ is defined as $\langle \mathbf{A}, \mathbf{B} \rangle \triangleq \text{tr}(\mathbf{A}^H \mathbf{B})$, where $\mathbf{A}^H$ is the conjugate transpose of $\mathbf{A}$, and $\text{tr}(\cdot)$ denotes the trace function. Likewise, the inner product of $\mathcal{A}$ and $\mathcal{B}$ in $\mathbb{C}^{n_1 \times n_2 \times n_3}$ is defined as $\langle \mathcal{A}, \mathcal{B} \rangle \triangleq \sum_{i=1}^{n_3} \langle \mathbf{A}^{(i)}, \mathbf{B}^{(i)} \rangle$. Let $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$, using the MATLAB notation, we define $\tilde{\mathcal{A}} \triangleq \text{fft}(\mathcal{A}, [], 3)$, which is the discrete Fourier transformation of $\mathcal{A}$ along the third dimension, and the inverse fft function is denoted as $\mathcal{A} \triangleq \text{ifft}(\tilde{\mathcal{A}}, [], 3)$. For a quaternion tensor $\dot{\mathcal{A}} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$, the ℓ_1 -norm is defined as $\|\dot{\mathcal{A}}\|_1 \triangleq \sum_{i,j,k} |\dot{\mathcal{A}}_{ijk}|$, the infinity norm is defined as $\|\dot{\mathcal{A}}\|_\infty \triangleq \max_{i,j,k} |\dot{\mathcal{A}}_{ijk}|$, and the Frobenius norm is denoted as $\|\dot{\mathcal{A}}\|_F \triangleq \sqrt{\sum_{i,j,k} |\dot{\mathcal{A}}_{ijk}|^2}$. The quaternion matrix nuclear norm is defined as $\|\dot{\mathbf{A}}\|_* \triangleq \sum_i \sigma_i(\dot{\mathbf{A}})$, where $\sigma_i(\dot{\mathbf{A}})$ is the i th singular value of $\dot{\mathbf{A}}$. The trace of $\dot{\mathcal{A}}$ is defined as $\text{tr}(\dot{\mathcal{A}}) \triangleq \sum_{i=1}^{n_3} \text{tr}(\dot{\mathbf{A}}^{(i)})$. We use the command symbols “qfft” and “iqfft” in MATLAB to represent quaternion fast Fourier transform and inverse quaternion fast Fourier transform.

2.3. Preliminaries

In the following, we will first review some basic definitions for the complex t-SVD. We introduce the definitions of the block diagonal matrix and the block circulant matrix of tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$ [32,33]. The block diagonal matrix of tensor $\mathcal{A}$ is defined as

$$\text{bdiag}(\mathcal{A}) \triangleq \text{diag}(\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(n_3)}) = \begin{pmatrix} \mathbf{A}^{(1)} & & & \\ & \mathbf{A}^{(2)} & & \\ & & \ddots & \\ & & & \mathbf{A}^{(n_3)} \end{pmatrix}, \quad (2)$$

where each frontal slice $\mathbf{A}^{(i)}$ of $\mathcal{A}$ lies on the diagonal in order. The block circulant matrix of tensor $\mathcal{A}$ is defined as

$$\text{bcirc}(\mathcal{A}) \triangleq \begin{pmatrix} \mathbf{A}^{(1)} & \mathbf{A}^{(n_3)} & \dots & \mathbf{A}^{(2)} \\ \mathbf{A}^{(2)} & \mathbf{A}^{(1)} & \dots & \mathbf{A}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}^{(n_3)} & \mathbf{A}^{(n_3-1)} & \dots & \mathbf{A}^{(1)} \end{pmatrix}. \quad (3)$$

Besides, a pair of folding operators are defined as follows

$$\text{unfold}(\mathcal{A}) \triangleq \begin{pmatrix} \mathbf{A}^{(1)} \\ \mathbf{A}^{(2)} \\ \vdots \\ \mathbf{A}^{(n_3)} \end{pmatrix}, \quad \text{fold}(\text{unfold}(\mathcal{A})) \triangleq \mathcal{A}. \quad (4)$$

Definition 1 (t-product [32,33]). Given complex tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times m}$ and $\mathcal{B} \in \mathbb{C}^{n_2 \times n_3 \times m}$, then the t-product $\mathcal{A} \star \mathcal{B}$ is a complex tensor of $n_1 \times n_3 \times m$, that is

$$\mathcal{A} \star \mathcal{B} = \text{fold}(\text{bcirc}(\mathcal{A})\text{unfold}(\mathcal{B})). \quad (5)$$

We note that using the proposed block-diagonalization results in [43], in the quaternion field, we can validly describe the t-product as the multiplication of a block quaternion circulant matrix with a block quaternion matrix column. This can also be interpreted as applying the quaternion Fourier transform to the tubes of the third mode of a quaternion tensor.

Definition 2 (Identity tensor [32,33]). We call a tensor $\mathcal{I} \in \mathbb{C}^{n \times n \times m}$ an identity tensor if its first frontal slice is the $n \times n$ identity matrix, and all other frontal slices are all zeros.

Definition 3 (Conjugate transpose [32,33]). Given a complex tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times m}$, then its conjugate transpose $\mathcal{A}^H \in \mathbb{C}^{n_2 \times n_1 \times m}$ is a tensor obtained by conjugate transposing each of the frontal slices and then reversing the order of transposed frontal slices from 2 through m .

Definition 4 (Unitary tensor [32,33]). A tensor $\mathcal{Q} \in \mathbb{C}^{n \times n \times m}$ is unitary if $\mathcal{Q}^H \star \mathcal{Q} = \mathcal{Q} \star \mathcal{Q}^H = \mathcal{I}$.

Definition 5 (f-diagonal tensor [32,33]). A tensor $\mathcal{A}$ is called f-diagonal if each frontal slice $\mathcal{A}^{(i)}$ is a diagonal matrix.

It has been proven that the completion performance based on t-SVD outperforms that based on other decompositions [32–34]. For quaternion tensor completion, using the t-SVD, the previous study used transform-based tensor product to obtain corresponding quaternion tensor singular value decomposition [25]. Nevertheless, this transform-based tensor product greatly increases the complexity of the tensor completion problem. Moreover, Pan et al. [43] have proved that a quaternion circulant matrix can be block-diagonalized into 1×1 and 2×2 blocks via the permuted discrete quaternion Fourier transform matrix. With such a block-diagonalized form, we first introduce the definitions of quaternion matrix and quaternion tensor, then the fast quaternion tensor singular value decomposition of t-product is introduced.

Definition 6 (Quaternion matrix [15]). The quaternion matrix is defined as $\hat{\mathbf{Q}} = (\hat{q}_{uv}) \in \mathbb{H}^{n_1 \times n_2}$, where each entry is a quaternion. It can be formulated as $\hat{\mathbf{Q}} = \mathbf{Q}_0 + \mathbf{Q}_1 \mathbf{i} + \mathbf{Q}_2 \mathbf{j} + \mathbf{Q}_3 \mathbf{k}$, where $\mathbf{Q}_l \in \mathbb{R}^{n_1 \times n_2}$ ($l = 0, 1, 2, 3$). Specifically, we call $\hat{\mathbf{Q}}$ is a pure quaternion matrix when $\mathbf{Q}_0 = \mathbf{0}$.

Definition 7 (Quaternion tensor [16]). The quaternion tensor is defined as $\hat{\mathcal{A}} = (\hat{a}_{i_1 i_2 \dots i_N}) \in \mathbb{H}^{n_1 \times n_2 \times \dots \times n_N}$, where each entry is a quaternion. It can be formulated as $\hat{\mathcal{A}} = \mathcal{A}_0 + \mathcal{A}_1 \mathbf{i} + \mathcal{A}_2 \mathbf{j} + \mathcal{A}_3 \mathbf{k}$, where $\mathcal{A}_l \in \mathbb{R}^{n_1 \times n_2 \times \dots \times n_N}$ ($l = 0, 1, 2, 3$). We call $\hat{\mathcal{A}}$ is a pure quaternion tensor when $\mathcal{A}_0$ is a zero tensor.

Theorem 1 (Fast Quaternion Tensor Singular Value Decomposition [43]). Given any quaternion tensor $\hat{\mathcal{A}} \in \mathbb{H}^{n_1 \times n_2 \times m}$, then there exist unitary quaternion tensors $\hat{\mathcal{U}} \in \mathbb{H}^{n_1 \times n_1 \times m}$ and $\hat{\mathcal{V}} \in \mathbb{H}^{n_2 \times n_2 \times m}$, and a f-diagonal tensor $\hat{\mathcal{S}} \in \mathbb{H}^{n_1 \times n_2 \times m}$ such that

$$\hat{\mathcal{A}} = \hat{\mathcal{U}} \star \hat{\mathcal{S}} \star \hat{\mathcal{V}}^H. \quad (6)$$

The decomposition (6) is called the fast quaternion tensor singular value decomposition (FQT-SVD) of $\hat{\mathcal{A}}$.

Fig. 2 offers a clear depiction of the three-dimensional FQT-SVD, where the quaternion tubal rank can be considered an extension of the real-valued tensor tubal rank.

Remark 1. According to [43], it is proven that a quaternion circulant matrix can be block-diagonalized into 1×1 and 2×2 blocks via the permuted discrete quaternion Fourier transform matrix, and we can make

use of this block-diagonalized form to study quaternion tensor singular value decomposition of a quaternion tensor. In more detail, given a quaternion tensor $\hat{\mathcal{A}} \in \mathbb{H}^{n_1 \times n_2 \times m}$, we can transform $\text{bcirc}(\hat{\mathcal{A}})$ into the Fourier domain by using the quaternion discrete Fourier matrix $\hat{\mathbf{F}}_\mu$,

$$(\hat{\mathbf{F}}_\mu \otimes \mathbf{I}_{n_1}) \text{bcirc}(\hat{\mathcal{A}}) (\hat{\mathbf{F}}_\mu^H \otimes \mathbf{I}_{n_2}) = \hat{\mathbf{D}}, \quad (7)$$

where $\otimes$ denotes the Kronecker product, the structure of $\hat{\mathbf{D}}$ is given as follows:

$$\hat{\mathbf{D}} = \begin{pmatrix} \hat{\mathbf{D}}_1 & & \\ & \hat{\mathbf{D}}_{1,2} & \hat{\mathbf{D}}_{2,2} \\ & \ddots & \ddots \\ & \hat{\mathbf{D}}_{1,m} & \hat{\mathbf{D}}_{2,m} \end{pmatrix}, \quad (8)$$

we remark that the size of $\hat{\mathbf{D}}_{1,t}$ and $\hat{\mathbf{D}}_{2,t}$ is $n_1 \times n_2$ ($t = 2, \dots, m$), the definitions of these can be referred to in [43]. Then, $\hat{\mathbf{D}}$ admits the following decomposition:

$$\hat{\mathbf{D}} = \hat{\mathbf{U}} \hat{\mathbf{\Sigma}} \hat{\mathbf{V}}^H, \quad (9)$$

where

$$\hat{\mathbf{\Sigma}} = \text{diag}(\hat{\mathbf{\Sigma}}_1, \hat{\mathbf{\Sigma}}_2, \dots, \hat{\mathbf{\Sigma}}_m),$$

$$\hat{\mathbf{U}} = \begin{pmatrix} \hat{\mathbf{U}}_1 & & \\ & \hat{\mathbf{U}}_{1,2} & \hat{\mathbf{U}}_{2,2} \\ & \ddots & \ddots \\ & \hat{\mathbf{U}}_{1,m} & \hat{\mathbf{U}}_{2,m} \end{pmatrix}, \hat{\mathbf{V}} = \begin{pmatrix} \hat{\mathbf{V}}_1 & & \\ & \hat{\mathbf{V}}_{1,2} & \hat{\mathbf{V}}_{2,2} \\ & \ddots & \ddots \\ & \hat{\mathbf{V}}_{1,m} & \hat{\mathbf{V}}_{2,m} \end{pmatrix}. \quad \text{Given}$$

$$\hat{\mathcal{L}} = \text{fold} \begin{pmatrix} \hat{\mathbf{U}}_1 \\ \hat{\mathbf{U}}_{1,2} + \hat{\mathbf{U}}_{2,2} \\ \vdots \\ \hat{\mathbf{U}}_{1,m} + \hat{\mathbf{U}}_{2,m} \end{pmatrix}, \hat{\mathcal{W}} = \text{fold} \begin{pmatrix} \hat{\mathbf{\Sigma}}_1 \\ \hat{\mathbf{\Sigma}}_2 \\ \vdots \\ \hat{\mathbf{\Sigma}}_m \end{pmatrix}, \quad (10)$$

$$\hat{\mathcal{R}} = \text{fold} \begin{pmatrix} \hat{\mathbf{V}}_1 \\ \hat{\mathbf{V}}_{1,2} + \hat{\mathbf{V}}_{2,2} \\ \vdots \\ \hat{\mathbf{V}}_{1,m} + \hat{\mathbf{V}}_{2,m} \end{pmatrix},$$

we note that the frontal slices of tensors $\hat{\mathcal{L}}$, $\hat{\mathcal{W}}$, and $\hat{\mathcal{R}}$ are computed by applying Quaternion Fast Fourier Transform (QFFT) along each tube of $\hat{\mathcal{U}}$, $\hat{\mathcal{S}}$, $\hat{\mathcal{V}}$, respectively. Thus, $\hat{\mathcal{U}}$, $\hat{\mathcal{S}}$, $\hat{\mathcal{V}}$ can be obtained by applying inverse QFFT along each tube of $\hat{\mathcal{L}}$, $\hat{\mathcal{W}}$, and $\hat{\mathcal{R}}$, respectively.

As mentioned previously that the tubal nuclear norm (Tubal-NN) is a convex relaxation of the ℓ_1 -norm associated with tensor rank [34]. It is expected to extend tensor nuclear norm to the quaternion field. From Theorem 1, in order to further propose the quaternion tensor truncated nuclear norm, we first given the definition of quaternion tensor nuclear norm for a three-dimensional quaternion tensor as follows.

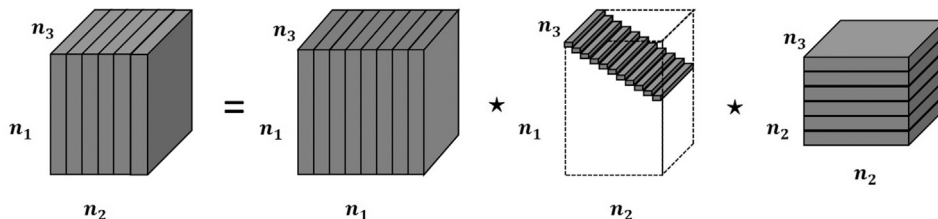


Fig. 2. The illustration of the FQT-SVD of a $n_1 \times n_2 \times n_3$ quaternion tensor.

Definition 8 (Quaternion Tensor Tubal Rank and Nuclear Norm). For a quaternion tensor $\mathcal{A} \in \mathbb{H}^{n_1 \times n_2 \times m}$, its FQT-SVD is given as $\mathcal{U} \star \mathcal{S} \star \mathcal{V}^H$. The quaternion tensor tubal rank of $\mathcal{A}$ is defined as the maximum rank among all frontal slices of the f-diagonal $\mathcal{S}$, i.e., $\text{rank}(\mathcal{A}) = \max_i \{\text{rank}(\mathcal{S}^{(i)})\}$. The tensor nuclear norm $\|\mathcal{A}\|_*$ is defined as the sum of singular values of all frontal slices of $\mathcal{S}$, i.e.,

$$\|\mathcal{A}\|_* \triangleq \text{tr}(\mathcal{S}) = \sum_{i=1}^m \text{tr}(\mathcal{S}^{(i)}), \quad (11)$$

note that $\tilde{\mathcal{S}}^{(1)} = \sum_{i=1}^m (\mathcal{S}^{(i)})$ and $\tilde{\mathcal{A}}^{(i)} = \tilde{\mathcal{U}}^{(i)} \tilde{\mathcal{S}}^{(i)} \tilde{\mathcal{V}}^{(i)H}$, such that

$$\|\mathcal{A}\|_* = \text{tr}(\tilde{\mathcal{S}}^{(1)}) = \|\tilde{\mathcal{A}}^{(1)}\|_*. \quad (12)$$

To better promote the low-rankness of the tensor, note that the larger the singular value, the less it will affect the rank, Hu et al. [36] further proposed the truncated nuclear norm regularization (TNNR) as a more accurate and much tighter substitute to the rank function. It inspired various works, and subsequently improved the efficacy of image or video completion [25,37–39]. Furthermore, we propose the quaternion tensor truncated nuclear norm as follows.

Definition 9 (Quaternion Tensor Truncated Nuclear Norm). Given a quaternion tensor $\mathcal{A} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$, the quaternion tensor truncated nuclear norm (QT-TNN) is defined as follows

$$\|\mathcal{A}\|_r \triangleq \|\tilde{\mathcal{A}}^{(1)}\|_r, \quad (13)$$

where $\|\tilde{\mathcal{A}}^{(1)}\|_r = \sum_{j=r+1}^{\min(n_1, n_2)} \sigma_j(\tilde{\mathcal{A}}^{(1)})$.

Lemma 1. Given a quaternion tensor $\mathcal{A} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$, the QT-TNN of $\mathcal{A}$ can be converted to

$$\|\mathcal{A}\|_r = \|\mathcal{A}\|_* - \sum_{j=1}^r \sigma_j(\tilde{\mathcal{A}}^{(1)}). \quad (14)$$

Proof. From (12) and (13), we have

$$\begin{aligned} \|\mathcal{A}\|_r &= \|\tilde{\mathcal{A}}^{(1)}\|_r \\ &= \sum_{j=r+1}^{\min(n_1, n_2)} \sigma_j(\tilde{\mathcal{A}}^{(1)}) \\ &= \sum_{j=1}^{\min(n_1, n_2)} \sigma_j(\tilde{\mathcal{A}}^{(1)}) - \sum_{j=1}^r \sigma_j(\tilde{\mathcal{A}}^{(1)}) \\ &= \|\tilde{\mathcal{A}}^{(1)}\|_* - \sum_{j=1}^r \sigma_j(\tilde{\mathcal{A}}^{(1)}) \\ &= \|\mathcal{A}\|_* - \sum_{j=1}^r \sigma_j(\tilde{\mathcal{A}}^{(1)}). \quad \square \end{aligned}$$

Definition 10 (Quaternion Singular Value Thresholding). Given a quaternion tensor $\mathcal{A} \in \mathbb{H}^{n_1 \times n_2 \times m}$, its FQT-SVD is given as $\mathcal{U} \star \mathcal{S} \star \mathcal{V}^H$. We denote the quaternion singular value thresholding (QSVT) operator as $\mathcal{D}_\tau$, is applied on each frontal slice of the f-diagonal tensor $\tilde{\mathcal{S}}$,

$$\begin{aligned} \mathcal{D}_\tau(\mathcal{A}) &\triangleq \mathcal{U} \star \mathcal{D}_\tau(\mathcal{S}) \star \mathcal{V}^H, \quad \mathcal{D}_\tau(\mathcal{S}) \triangleq \text{iqfft}(\mathcal{D}_\tau(\tilde{\mathcal{S}})), \\ \mathcal{D}_\tau(\tilde{\mathcal{S}}^{(i)}) &\triangleq \text{diag}(\max\{\sigma_i - \tau, 0\}_{1 \leq i \leq r}), i = 1, 2, \dots, m. \end{aligned} \quad (15)$$

3. Quaternion tensor truncated nuclear norm regularization

3.1. Problem formulation

In this section, we will begin by reviewing the theory of quaternion tensor completion. Then, a novel quaternion-based tensor completion model is proposed.

Low-rank quaternion tensor completion aims to employ the low-rankness as previously described to establish the relationship between the observed and missing elements in the target quaternion tensor. Mathematically, the optimization model for quaternion tensor completion problem can be formulated as:

$$\begin{aligned} \min_{\mathcal{T}} \quad & \text{rank}(\mathcal{T}) \\ \text{s.t.} \quad & \mathcal{T}_\Omega = \mathcal{M}_\Omega, \end{aligned} \quad (16)$$

where $\mathcal{T}$ is the underlying quaternion tensor, while $\mathcal{M}$ represents the initial incomplete data, and Ω denotes the set of positions corresponding to the known elements.

Inspired by the success of FQT-SVD and TNNR methods, we propose the quaternion tensor truncated nuclear norm regularization (QT-TNNR) model as follows

$$\begin{aligned} \min_{\mathcal{T}} \quad & \|\mathcal{T}\|_r \\ \text{s.t.} \quad & \mathcal{T}_\Omega = \mathcal{M}_\Omega, \end{aligned} \quad (17)$$

From Lemma 1 and Theorem 3.1 in [36], the model (17) can be rewritten as

$$\begin{aligned} \min_{\mathcal{T}} \quad & \|\mathcal{T}\|_* - \max_{\substack{\tilde{\mathcal{A}}^{(1)} \tilde{\mathcal{A}}^{(1)H} = \mathbf{I}, \\ \tilde{\mathcal{B}}^{(1)} \tilde{\mathcal{B}}^{(1)H} = \mathbf{I}}} \text{tr}(\tilde{\mathcal{A}}^{(1)} \tilde{\mathcal{T}}^{(1)} \tilde{\mathcal{B}}^{(1)H}) \\ \text{s.t.} \quad & \mathcal{T}_\Omega = \mathcal{M}_\Omega. \end{aligned} \quad (18)$$

Note that

$$\text{tr}(\mathcal{A} \star \mathcal{T} \star \mathcal{B}^H) = \text{tr}(\tilde{\mathcal{A}}^{(1)} \tilde{\mathcal{T}}^{(1)} \tilde{\mathcal{B}}^{(1)H}), \quad (19)$$

the proof is provided in Appendix A, (18) can be rewritten as

$$\begin{aligned} \min_{\mathcal{T}} \quad & \|\mathcal{T}\|_* - \max_{\substack{\mathcal{A} \star \mathcal{A}^H = \mathbf{I}, \\ \mathcal{B} \star \mathcal{B}^H = \mathbf{I}}} \text{tr}(\mathcal{A} \star \mathcal{T} \star \mathcal{B}^H) \\ \text{s.t.} \quad & \mathcal{T}_\Omega = \mathcal{M}_\Omega. \end{aligned} \quad (20)$$

To solve (20), in the ℓ th iteration, we consider to divide the optimization into two steps. First, we solve the following problem by fixing $\mathcal{T}_\ell$,

$$\max_{\substack{\mathcal{A}_\ell \star \mathcal{A}_\ell^H = \mathbf{I}, \\ \mathcal{B}_\ell \star \mathcal{B}_\ell^H = \mathbf{I}}} \text{tr}(\mathcal{A}_\ell \star \mathcal{T}_\ell \star \mathcal{B}_\ell^H). \quad (21)$$

According to the Remark 1, suppose $\mathcal{T}_\ell = \mathcal{U}_\ell \star \mathcal{S}_\ell \star \mathcal{V}_\ell^H$, such that $\mathcal{A}_\ell$ and $\mathcal{B}_\ell$ in (21) are given as follows

$$\mathcal{A}_\ell = \mathcal{U}_\ell(:, 1:r, :)^H, \quad \mathcal{B}_\ell = \mathcal{V}_\ell(:, 1:r, :)^H, \quad (22)$$

where $\mathcal{U}_\ell(:, 1:r, :)$ and $\mathcal{V}_\ell(:, 1:r, :)$ define the operator of selecting the first r columns in the second dimension of $\mathcal{U}_\ell$ and $\mathcal{V}_\ell$ (using MATLAB notation).

Next, we can compute $\mathcal{T}_{\ell+1}$ by fixing $\mathcal{A}_\ell$ and $\mathcal{B}_\ell$ through the following problem:

$$\begin{aligned} \min_{\mathcal{T}} \quad & \|\mathcal{T}\|_* - \text{tr}(\mathcal{A}_\ell \star \mathcal{T} \star \mathcal{B}_\ell^H) \\ \text{s.t.} \quad & \mathcal{T}_\Omega = \mathcal{M}_\Omega, \end{aligned} \quad (23)$$

In the next subsection, we will introduce how to solve (23) clearly. The optimization converges to a local minimum of (20) by alternately taking two steps above. The structure of the proposed approach is summarized in Algorithm 1.

Algorithm 1 Low-rank Quaternion Tensor Completion by Truncated Nuclear Norm Regularization (QT-TNNR).

Input: The observed incomplete quaternion tensor $\dot{\mathcal{M}} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$; Ω , the index set of the observed elements; $\varepsilon = 10^{-3}$, $l_{\max} = 500$.
Initialize : $\dot{\mathcal{T}}_1 = \dot{\mathcal{M}}_{\Omega}$, $\ell = 1$.

1: **Repeat**

2: **Step 1:** Given $\dot{\mathcal{T}}_{\ell} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$,

$$[\dot{\mathcal{U}}_{\ell}, \dot{\mathcal{S}}_{\ell}, \dot{\mathcal{V}}_{\ell}] = \text{FQT-SVD}(\dot{\mathcal{T}}_{\ell}),$$

where the orthogonal quaternion tensors are

$$\dot{\mathcal{U}}_{\ell} \in \mathbb{H}^{n_1 \times n_1 \times n_3}, \quad \dot{\mathcal{V}}_{\ell} \in \mathbb{H}^{n_2 \times n_2 \times n_3}.$$

3: Calculating $\dot{\mathcal{A}}_{\ell}$ and $\dot{\mathcal{B}}_{\ell}$ as follows ($r \leq \min\{n_1, n_2\}$):

$$\dot{\mathcal{A}}_{\ell} = \dot{\mathcal{U}}_{\ell}(:, 1:r, :)^H, \quad \dot{\mathcal{B}}_{\ell} = \dot{\mathcal{V}}_{\ell}(:, 1:r, :)^H.$$

4: **Step 2:** Solving the optimization problem as follows

$$\dot{\mathcal{T}}_{\ell+1} = \arg \min_{\dot{\mathcal{T}}} \|\dot{\mathcal{T}}\|_* - \text{tr}(\dot{\mathcal{A}}_{\ell} \star \dot{\mathcal{T}} \star \dot{\mathcal{B}}_{\ell}^H) \quad \text{s.t.} \quad \dot{\mathcal{T}}_{\Omega} = \dot{\mathcal{M}}_{\Omega}.$$

5: **Until** $\|\dot{\mathcal{T}}_{\ell+1} - \dot{\mathcal{T}}_{\ell}\|_F \leq \varepsilon$ or $\ell > L$, $\dot{\mathcal{T}}_{\text{opt}} = \dot{\mathcal{T}}_{\ell+1}$.

Output: the recovered quaternion tensor $\dot{\mathcal{T}}_{\text{opt}}$.

3.2. Optimization procedure

In this section, we show how to solve the optimization problem (23). We know that the ADMM is widely adopted in solving constrained optimization problems because of the convergence guarantee in polynomial time [4,16,25]. Boyd et al. [45] systematically introduced the theoretical properties of ADMM and its advantages in large-scale distributed optimization, promoting the method as a core tool in machine learning and engineering fields. For efficiency, we used it in Step 2 in Algorithm 1. We note that by introducing an auxiliary variable $\dot{\mathcal{H}}$ to relax the objective, (23) can be reformulated as the following problem:

$$\begin{aligned} \min_{\dot{\mathcal{T}}, \dot{\mathcal{H}}} \quad & \|\dot{\mathcal{T}}\|_* - \text{tr}(\dot{\mathcal{A}}_{\ell} \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_{\ell}^H) \\ \text{s.t.} \quad & \dot{\mathcal{T}} = \dot{\mathcal{H}}, \quad \dot{\mathcal{H}}_{\Omega} = \dot{\mathcal{M}}_{\Omega}. \end{aligned} \quad (24)$$

Inspired by the ADMM framework utilized in the quaternion field [46], this problem can be solved by the ADMM framework, which minimizes the following augmented Lagrangian function:

$$L(\dot{\mathcal{T}}, \dot{\mathcal{H}}, \dot{\mathcal{Y}}) = \|\dot{\mathcal{T}}\|_* - \text{tr}(\dot{\mathcal{A}}_{\ell} \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_{\ell}^H) + \langle \dot{\mathcal{Y}}, \dot{\mathcal{T}} - \dot{\mathcal{H}} \rangle + \frac{\mu}{2} \|\dot{\mathcal{T}} - \dot{\mathcal{H}}\|_F^2, \quad (25)$$

where $\dot{\mathcal{Y}}$ is a Lagrange multiplier and $\mu > 0$ is the scalar penalty parameter. We set $\dot{\mathcal{T}}_1 = \dot{\mathcal{M}}_{\Omega}$, $\dot{\mathcal{H}}_1 = \dot{\mathcal{T}}_1$, and $\dot{\mathcal{Y}}_1 = \dot{\mathcal{T}}_1$ as the initialization. A natural way to solve the problem is to iteratively optimize the augmented Lagrangian function (25) over one variable, while fixing the others. To update each variable, in the $(k+1)$ th iteration, we divide the optimization of (24) into the following three steps:

Step 1: Updating $\dot{\mathcal{T}}_{k+1}$ from $L(\dot{\mathcal{T}}, \dot{\mathcal{H}}_k, \dot{\mathcal{Y}}_k)$ by keeping $\dot{\mathcal{H}}_k$ and $\dot{\mathcal{Y}}_k$ invariant:

$$\begin{aligned} \dot{\mathcal{T}}_{k+1} &= \arg \min_{\dot{\mathcal{T}}} \|\dot{\mathcal{T}}\|_* + \langle \dot{\mathcal{Y}}_k, \dot{\mathcal{T}} - \dot{\mathcal{H}}_k \rangle + \frac{\mu}{2} \|\dot{\mathcal{T}} - \dot{\mathcal{H}}_k\|_F^2 \\ &= \arg \min_{\dot{\mathcal{T}}} \|\dot{\mathcal{T}}\|_* + \frac{\mu}{2} \|\dot{\mathcal{T}} - (\dot{\mathcal{H}}_k - \frac{1}{\mu} \dot{\mathcal{Y}}_k)\|_F^2. \end{aligned} \quad (26)$$

According to [47], for each $\tau > 0$ and $\dot{\mathcal{A}} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$, the QSVT operator obeys

$$D_{\tau}(\dot{\mathcal{A}}) = \arg \min_{\dot{\mathcal{T}}} \tau \|\dot{\mathcal{T}}\|_* + \frac{1}{2} \|\dot{\mathcal{T}} - \dot{\mathcal{A}}\|_F^2. \quad (27)$$

From (27) and the definition of QSVT operator in Definition 10, we can solve (26) efficiently by

$$\dot{\mathcal{T}}_{k+1} = D_{\frac{1}{\mu}} \left(\dot{\mathcal{H}}_k - \frac{1}{\mu} \dot{\mathcal{Y}}_k \right). \quad (28)$$

Step 2: Updating $\dot{\mathcal{H}}_{k+1}$ by fixing $\dot{\mathcal{T}}_{k+1}$ and $\dot{\mathcal{Y}}_k$ invariant:

$$\begin{aligned} \dot{\mathcal{H}}_{k+1} &= \arg \min_{\dot{\mathcal{H}}} L(\dot{\mathcal{T}}_{k+1}, \dot{\mathcal{H}}, \dot{\mathcal{Y}}_k) \\ &= \arg \min_{\dot{\mathcal{H}}} \frac{\mu}{2} \|\dot{\mathcal{T}}_{k+1} - \dot{\mathcal{H}}\|_F^2 - \text{tr}(\dot{\mathcal{A}}_{\ell} \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_{\ell}^H) + \langle \dot{\mathcal{Y}}_k, \dot{\mathcal{T}}_{k+1} - \dot{\mathcal{H}} \rangle. \end{aligned} \quad (29)$$

Clearly, (29) is quadratic in relation to $\dot{\mathcal{H}}$. Therefore, we update the target variables by solving the Karush-Kuhn-Tucker (KKT) conditions. By setting the derivative of (29) to zero, i.e., let $\frac{\partial L(\dot{\mathcal{T}}_{k+1}, \dot{\mathcal{H}}, \dot{\mathcal{Y}}_k)}{\partial \dot{\mathcal{H}}} = 0$, we can derive the closed-form solution as follows

$$\begin{aligned} \dot{\mathcal{H}}_{k+1}(\cdot, \cdot, i) &= \dot{\mathcal{T}}_{k+1}(\cdot, \cdot, i) \\ &+ \frac{1}{\mu} \left(\left(\sum_{i=1}^{n_3} \dot{\mathcal{A}}_{\ell}(\cdot, \cdot, i) \right)^H \left(\sum_{i=1}^{n_3} \dot{\mathcal{B}}_{\ell}(\cdot, \cdot, i) \right) + \dot{\mathcal{Y}}_k(\cdot, \cdot, i) \right). \end{aligned} \quad (30)$$

For the mathematical derivation of $\text{tr}(\dot{\mathcal{A}}_{\ell} \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_{\ell}^H)$ in (29), please refer to Appendix B. Additionally, we note that the values of all observed elements should be constant in each iteration, i.e.,

$$\dot{\mathcal{H}}_{k+1} = (\dot{\mathcal{H}}_{k+1})_{\Omega^c} + \dot{\mathcal{M}}_{\Omega}. \quad (31)$$

Step 3: Updating $\dot{\mathcal{Y}}_{k+1}$ directly through

$$\dot{\mathcal{Y}}_{k+1} = \dot{\mathcal{Y}}_k + \mu(\dot{\mathcal{T}}_{k+1} - \dot{\mathcal{H}}_{k+1}). \quad (32)$$

3.3. The computational complexity analysis

In this section, we analyze the computational complexity of Algorithm 1. For simplicity, assume $n_1 = n_2 = n_3 = n$. The per-iteration complexity primarily stems from solving (21) and (24). Optimizing (21): Each iteration costs $O(n^4)$ (dominated by n^4). Optimizing (24) via ADMM: Each iteration involves (28), (30), (31), and (32), with a cost of $O(n^4)$. In Algorithm 1, let ℓ be the number of iterations for Steps 1–2, the computational complexity of (21) is $O(\ell n^4)$, assume k is the number of iterations for (24), the computational complexity of (24) is $O(\ell k n^4)$, then the total complexity is $O(\ell n^4 + \ell k n^4)$.

4. Experimental results

In this section, experiments on six natural color videos are conducted to evaluate the performance of the proposed QT-TNNR method. We compare them with several existing state-of-the-art approaches, including T-TNN [39], LRC-QT [16], QT-RNNS [25], the deep learning-based approach 3DCN [48] and 2DCN [49].

In the following, a video contains n_3 frames where each frame is a $n_1 \times n_2$ RGB image. The color value of each pixel is encoded in a pure quaternion, that is, a pixel value at location (s, p) is given by $m_{sp} = \mathbf{R}_{sp}\mathbf{i} + \mathbf{G}_{sp}\mathbf{j} + \mathbf{B}_{sp}\mathbf{k}$, where $\mathbf{R}$, $\mathbf{G}$, $\mathbf{B}$ denote the red, green and blue components of each pixel, respectively. Hence, the video data can be represented by a pure quaternion tensor $\dot{\mathcal{A}} \in \mathbb{H}^{n_1 \times n_2 \times n_3}$.

4.1. Experimental settings

Test Data and Experimental Facility: We use the color video dataset: YUV Video Sequence,¹ where each sequence contains at least 150 frames. Six color videos are selected, including Bus, News, Silent, Suzie,

¹ <http://trace.eas.asu.edu/yuv/>

Table 1
SI and TI values for all videos.

Videos	Spatial information (SI)	Temporal information (TI)
Bus	0.59	0.12
News	0.58	0.10
Silent	0.33	0.04
Suzie	0.23	0.06
Container	0.50	0.01
Mobile	183.41	18.07

Container, and Mobile. We adopt the first 20 frames in the following experiments, such that the first video is of size $288 \times 352 \times 3 \times 20$ and the last five videos are of size $144 \times 176 \times 3 \times 20$. Then we reconstruct them as pure quaternion tensors with sizes $288 \times 352 \times 20$ and $144 \times 176 \times 20$, respectively. The Sampling Ratio (SR) is set to 50 %, 40 %, 30 %, 20 %, 10 %, respectively. We analyze the Spatial Information (SI) and Temporal Information (TI) [50] characteristics of these videos. SI denotes the scale of the number of spatial details in a picture. The larger the value of SI, the higher the spatial complexity. TI denotes a measure of the amount of temporal variation in a video sequence. The larger the value of TI, the higher the motion of the video sequence. The values of SI and TI for each of the six videos are calculated according to the literature [50] and are shown in Table 1. As shown in Table 1, Mobile has higher SI and TI values than the other five video sets, i.e., Mobile has higher spatial complexity and higher motion. Significant variations exist in both SI and TI metrics across different datasets. Conducting experiments on these diverse datasets enables a comprehensive evaluation of algorithm performance across various scenarios. All the experiments are implemented in MATLAB R2023a, on a PC with a 1.30 GHz CPU and 32GB RAM.

Evaluation Indicators: Three indicators are used to assess the performance of the proposed method: peak signal-to-noise ratio (PSNR), feature similarity (FSIM), and structural similarity (SSIM), which are respectively defined as:

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{Peakval}^2}{\text{MSE}} \right),$$

where Peakval is taken from the range of the pixel value datatype (e.g., for uint8 pixel value, it is 255), MSE is the mean square error, i.e., $\text{MSE} = \|\mathcal{X} - \mathcal{T}\|_F^2 / \text{numel}(\mathcal{X})$, where $\mathcal{X}$, $\mathcal{T}$, and $\text{numel}(\mathcal{X})$ are the recovered, truth data, and the total number of missing elements in a tensor, respectively;

$$\text{FSIM} = \frac{\sum_{z \in \Delta} S_L(z) PC_m(z)}{\sum_{z \in \Delta} PC_m(z)},$$

where Δ denotes the whole image spatial domain. The $PC_{\mathcal{T}}(z)$ denotes the phase congruency for position z of image $\mathcal{T}$, then $PC_m(z) = \max \{PC_{\mathcal{T}}(z), PC_{\mathcal{X}}(z)\}$, $S_L(z)$ represents the gradient magnitude at position z ;

$$\text{SSIM} = \frac{(2\mu_{\mathcal{T}}\mu_{\mathcal{X}} + C_1)(2\sigma_{\mathcal{T}\mathcal{X}} + C_2)}{(\mu_{\mathcal{T}}^2 + \mu_{\mathcal{X}}^2 + C_1)(\sigma_{\mathcal{T}}^2 + \sigma_{\mathcal{X}}^2 + C_2)},$$

where $\mu_{\mathcal{T}}$, $\mu_{\mathcal{X}}$, $\sigma_{\mathcal{T}}$, $\sigma_{\mathcal{X}}$, and $\sigma_{\mathcal{T}\mathcal{X}}$ are the local means, standard deviations, and cross-covariance for images $\mathcal{T}$ and $\mathcal{X}$, $C_1 = (0.01 L)^2$, $C_2 = (0.03 L)^2$, L denotes the specified dynamic range of the pixel values.

Parameters Setting: In this paper, each algorithm concludes either when $\|\hat{\mathcal{T}}_{k+1} - \hat{\mathcal{T}}_k\|_F$ is sufficiently small or when the maximum number of iterations is reached. We denote $\hat{\mathcal{T}}_{opt}$ as the final output. We set $\varepsilon = 10^{-3}$ and $l_{max} = 500$ in our approach. To balance the efficiency and the accuracy of our method, we employed cross-validation on the training set to determine the optimal value of μ . The sensitivity analysis on six color videos is provided in Table 2, where “Iter” denotes the number of iterations and “Time(s)” denotes the running time in seconds. Each parameter of compared methods is readjusted to be optimal and the best results are reported.

Table 2

The sensitivity analysis of μ for six color videos at a sampling ratio (SR) of 50 %.

SR = 50 %						
Videos	μ	PSNR	FSIM	SSIM	Iter	Time (s)
Bus	0.01	27.2263	0.9923	0.7807	50	3469.69
	0.05	27.1652	0.9922	0.7762	55	3816.66
	0.10	27.2034	0.9922	0.7756	53	3677.87
	0.50	27.5434	0.9928	0.7927	59	4315.79
News	0.01	38.5073	0.9994	0.9764	42	439.92
	0.05	38.6793	0.9995	0.9783	29	303.75
	0.10	38.8848	0.9995	0.9791	21	219.96
	0.50	38.3929	0.9994	0.9769	52	497.63
Silent	0.01	38.8043	0.9991	0.9721	35	317.00
	0.05	38.9344	0.9992	0.9736	30	271.71
	0.10	39.2668	0.9992	0.9748	27	244.54
	0.50	38.7478	0.9992	0.9726	51	461.91
Suzie	0.01	36.7831	0.9982	0.9478	24	213.96
	0.05	36.7255	0.9982	0.9474	28	249.62
	0.10	37.3458	0.9984	0.9525	16	166.44
	0.50	36.6429	0.9981	0.9464	32	285.28
Container	0.01	39.7852	0.9992	0.9775	36	339.27
	0.05	39.4105	0.9992	0.9779	30	282.72
	0.10	39.9938	0.9992	0.9801	20	188.48
	0.50	39.0434	0.9991	0.9757	89	838.75
Mobile	0.0005	24.7158	0.8968	0.7742	21	234.26
	0.001	25.6538	0.9078	0.7980	13	143.79
	0.005	25.4567	0.9071	0.8097	124	1174.83
	0.01	25.4557	0.9071	0.8096	246	3749.15

4.2. Color video recovery results

4.2.1. Numerical results

In our experiments, the PSNR, FSIM, and SSIM of different methods on the six color videos with various SR are listed in Table 3. The best numerical results are bolded. Table 3 shows that, compared to other methods, QT-TNNR maintains the highest PSNR, FSIM, and SSIM values in most cases. It is mainly due to the efficiency of the truncation strategy of QT-TNNR and the advantage of quaternion representation of color pixel values.

Observation of Table 3 indicates that QT-TNNR demonstrates relatively superior performance across all three metrics evaluated by the comparative methods. Specifically, in the PSNR metric comparison, QT-TNNR exhibits exceptional recovery performance when the SR exceeds 30 %. At lower SR levels, T-TNN and the deep learning methods (3DCN and 2DCN) maintain better numerical results, with QT-TNNR following closely behind. This demonstrates QT-TNNR’s efficacy in pixel value recovery. Regarding the FSIM metric, QT-TNNR’s performance is particularly outstanding, achieving optimal results with only 20 % known information. However, at SR=10 %, 3DCN shows relatively superior performance, which can be attributed to its advantage in leveraging three-channel relationships for feature similarity extraction. In contrast, QT-TNNR’s excellence stems from both the intrinsic three-channel correlations inherent in quaternion representations and its effective low-rank recovery. These results confirm QT-TNNR’s capability to effectively preserve phase congruency and gradient features. For the SSIM metric, QT-TNNR also exhibits favorable performance, while T-TNN achieves relatively good results. This indicates that the truncated nuclear norm approach enhances model completion effectiveness, maintaining high recovery levels in luminance, contrast, and structural similarity. Overall, QT-TNNR demonstrates superior numerical performance. To visualize these findings more intuitively, Fig. 3 illustrates that QT-TNNR’s histograms for all three metrics are higher at both SR = 50 % and SR = 30 %. Conversely, at SR=10 %, the histograms of the deep learning methods (3DCN and 2DCN) are higher, with QT-TNNR and T-TNN following behind. In summary, QT-TNNR exhibits robust numerical performance.

Table 3

The PSNR, FSIM, and SSIM comparison of different methods on six color videos with various sampling ratios (SR).

Videos	SR	PSNR						FSIM						SSIM					
		QT-TNNR	T-TNN	LRC-QT	QT-RNNS	3DCN	2DCN	QT-TNNR	T-TNN	LRC-QT	QT-RNNS	3DCN	2DCN	QT-TNNR	T-TNN	LRC-QT	QT-RNNS	3DCN	2DCN
Bus	50 %	27.5434	25.1627	19.2591	23.7345	21.4301	11.5925	0.9928	0.9801	0.9577	0.9831	0.9794	0.8896	0.7927	0.7296	0.5576	0.7450	0.6689	0.1319
	40 %	24.6099	24.0686	19.0438	22.3294	20.4368	11.8763	0.9862	0.9739	0.9512	0.9785	0.9700	0.8957	0.6742	0.6892	0.5384	0.7075	0.6070	0.1312
	30 %	23.1505	22.8198	17.3431	21.2017	19.5083	12.6711	0.9790	0.9656	0.9385	0.9726	0.9598	0.8903	0.5899	0.6405	0.4801	0.6663	0.5359	0.1247
	20 %	21.1954	21.2707	15.0252	19.2324	18.9449	12.2057	0.9648	0.9529	0.9216	0.9585	0.9498	0.8929	0.4798	0.5814	0.4045	0.5848	0.4810	0.1528
	10 %	19.0854	19.2576	12.6776	15.4869	18.1267	12.5671	0.9377	0.9264	0.9041	0.9216	0.9413	0.8928	0.3602	0.4938	0.3294	0.4227	0.4105	0.1802
News	50 %	38.8848	33.5495	24.9054	20.2342	25.9991	25.6761	0.9995	0.9938	0.9886	0.9753	0.9929	0.9922	0.9791	0.9355	0.7508	0.6588	0.8566	0.8771
	40 %	34.8024	32.3317	21.3487	18.9932	26.7092	26.2031	0.9987	0.9918	0.9792	0.9680	0.9929	0.9928	0.9469	0.9202	0.6637	0.6129	0.8783	0.8779
	30 %	30.6929	30.8542	19.2147	17.8263	25.3966	26.9885	0.9948	0.9888	0.9698	0.9591	0.9905	0.9938	0.8470	0.8950	0.5852	0.5659	0.8278	0.8889
	20 %	25.9483	28.2351	17.7764	16.2252	23.9083	27.8945	0.9909	0.9832	0.9591	0.9433	0.9865	0.9947	0.7704	0.8497	0.4506	0.4955	0.7896	0.9014
	10 %	21.3495	23.9976	14.8239	13.4124	23.3876	28.8430	0.9739	0.9628	0.9361	0.9062	0.9859	0.9959	0.5806	0.6851	0.3418	0.3824	0.7939	0.9070
Silent	50 %	39.2668	32.8677	22.7121	21.2193	26.8406	25.3713	0.9992	0.9947	0.9735	0.9685	0.9915	0.9886	0.9748	0.9271	0.6120	0.6159	0.8370	0.8103
	40 %	35.1785	31.6385	18.9031	19.9706	26.2219	26.2199	0.9983	0.9932	0.9546	0.9600	0.9891	0.9911	0.9430	0.9100	0.5123	0.5751	0.8165	0.8232
	30 %	32.6339	30.0865	17.3644	18.5157	25.3909	26.7079	0.9970	0.9907	0.9439	0.9486	0.9864	0.9927	0.9096	0.8760	0.4006	0.5278	0.8015	0.8230
	20 %	29.1878	27.6374	15.0791	15.6449	24.3931	27.8578	0.9938	0.9846	0.9259	0.9220	0.9829	0.9948	0.8293	0.8176	0.3494	0.4190	0.7680	0.8417
	10 %	25.0735	18.8702	12.4816	11.5433	23.7366	29.0839	0.9818	0.9607	0.9033	0.8901	0.9796	0.9951	0.6567	0.6630	0.1868	0.3157	0.7550	0.8551
Suzie	50 %	37.3458	32.5949	23.1419	22.2337	26.3569	22.3672	0.9984	0.9917	0.9537	0.9510	0.9910	0.9762	0.9525	0.8944	0.5255	0.5660	0.7687	0.6135
	40 %	33.8992	31.7428	20.9735	20.8998	26.9801	23.1300	0.9964	0.9902	0.9347	0.9401	0.9868	0.9793	0.9072	0.8812	0.4659	0.5188	0.7675	0.6064
	30 %	31.1375	30.5516	19.1869	19.1846	25.8511	24.1668	0.9914	0.9872	0.9186	0.9238	0.9825	0.9814	0.8394	0.8605	0.3706	0.4574	0.7363	0.6418
	20 %	28.9076	28.6777	16.8515	16.5160	24.0131	25.2687	0.9902	0.9820	0.8976	0.8917	0.9768	0.9857	0.8038	0.8295	0.3029	0.3722	0.6865	0.6625
	10 %	25.2841	20.2411	12.4988	12.5756	24.3683	25.0123	0.9725	0.9675	0.8720	0.8630	0.9740	0.9778	0.6701	0.7262	0.2468	0.3010	0.6709	0.6858
Container	50 %	39.9938	34.9101	21.0316	18.3245	27.1598	25.0411	0.9992	0.9977	0.9603	0.9559	0.9965	0.9940	0.9801	0.9494	0.5475	0.5513	0.8910	0.8131
	40 %	36.0181	33.4957	19.0487	16.9664	26.7283	25.6375	0.9983	0.9973	0.9516	0.9481	0.9952	0.9949	0.9511	0.9391	0.4742	0.5077	0.8730	0.8267
	30 %	32.2323	31.6661	15.2367	15.3001	25.6044	25.7791	0.9959	0.9963	0.9361	0.9361	0.9939	0.9949	0.8942	0.9246	0.4099	0.4492	0.8477	0.8286
	20 %	28.1970	28.6196	14.1172	12.8947	24.9746	25.6099	0.9937	0.9935	0.9267	0.9152	0.9929	0.9954	0.8470	0.8899	0.3097	0.3752	0.8367	0.8140
	10 %	23.8283	18.1490	10.5520	9.4863	24.2357	26.2256	0.9810	0.9797	0.9072	0.8898	0.9904	0.9949	0.6991	0.7648	0.1504	0.3048	0.7648	0.8221
Mobile	50 %	25.6538	22.6962	15.3217	12.8918	17.6138	11.6331	0.9078	0.8676	0.7679	0.6850	0.8081	0.6071	0.7980	0.8355	0.5713	0.5471	0.6226	0.1234
	40 %	23.0411	17.9156	16.1752	11.1545	16.9402	12.0894	0.8668	0.7829	0.7567	0.6356	0.7762	0.6193	0.7177	0.7042	0.5464	0.4833	0.5595	0.1512
	30 %	20.7856	14.3060	12.7206	9.5578	16.1765	12.2545	0.8186	0.7033	0.7002	0.5923	0.7451	0.6216	0.6094	0.5700	0.3592	0.4239	0.4916	0.1649
	20 %	18.5534	11.4177	13.2255	8.1606	15.4674	12.3882	0.7527	0.6336	0.6699	0.5584	0.7133	0.6295	0.4748	0.4473	0.3076	0.3670	0.4106	0.2249
	10 %	16.1400	8.9061	10.7515	6.8271	13.8913	12.4807	0.6646	0.5747	0.5946	0.5192	0.6571	0.6119	0.3040	0.3403	0.1572	0.3068	0.2841	0.2302

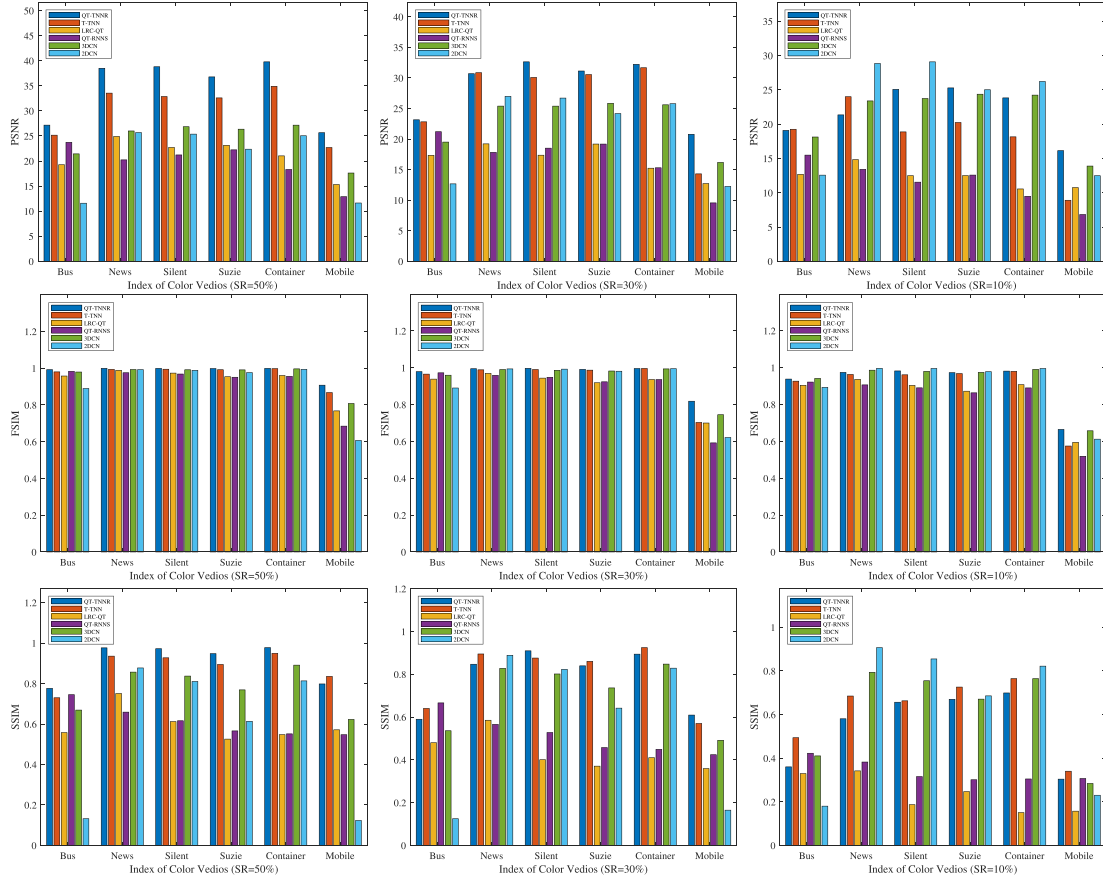
**Fig. 3.** The comparison of the PSNR, FSIM, and SSIM values of various methods for color video restoration.

Table 4

Comparison of iteration counts and computational times across various methods on six color videos at a sampling ratio (SR) of 50 %.

		SR = 50 %						
		Methods	QT-TNNR	T-TNN	LRC-QT	QT-RNNS	3DCN	2DCN
Bus	Iter	50	19	17	16	1000	500	
	Time (s)	3469.69	452.18	339.35	3280.55	1376.82	2197.65	
News	Iter	42	17	18	19	1000	1000	
	Time (s)	439.92	174.34	130.84	1502.01	353.08	542.02	
Silent	Iter	35	18	21	20	1000	1000	
	Time (s)	317.00	175.08	152.65	1581.64	365.95	1227.19	
Suzie	Iter	24	13	19	20	1000	1000	
	Time (s)	213.96	120.68	138.11	1487.32	439.84	986.35	
Container	Iter	36	16	23	22	1000	500	
	Time (s)	339.27	153.65	167.18	1739.17	398.78	527.95	
Mobile	Iter	13	50	23	8	1000	500	
	Time	143.79	33.40	37.01	220.14	399.60	783.18	

It is noteworthy that discrepancies among these three metrics are observed across different methodologies, particularly between the deep learning-based 3DCN and 2DCN approaches. These discrepancies primarily stem from the distinct mechanisms between tensor completion-based methods and deep learning-based approaches in image restoration, such as differences in blurring mechanisms and inpainting strategies. Therefore, our method demonstrates comprehensive superiority in both pixel recovery and structural restoration. While 3DCN and 2DCN achieve satisfactory structural restoration, they exhibit limitations in pixel-level accuracy, highlighting the inherent advantages of quaternion-based approaches in processing color images. Furthermore, Table 4 gives the number of iteration steps and running time of the different comparison methods when SR=50 %. As shown in Tables 3 and 4, our method achieves competitive iteration counts and computational time while maintaining absolute superiority in PSNR values. In low information scenarios, deep learning methods achieve better numerical metrics but notably introduce ghosting artifacts from training data patterns. By contrast, QT-TNNR maintains visually natural restoration, attributable to its constrained low-rank space, avoiding overfitting while preserving structural coherence. Specifically, the textural recovery capabilities of each method will be further analyzed in the following section.

4.2.2. Visualization results

Figs. 4–9 show the recovery results by different approaches for the 1st, 7th, 14th and 20th frames of Bus video with SR = 50 %, News video with SR = 40 %, Silent video with SR = 30 %, Suzie video with SR = 20 %, Container video with SR = 10 %, and Mobile video with SR=50 %, respectively. For the 1st, 7th, 14th and 20th frames of Bus video with SR = 50 %, News video with SR = 40 %, and Silent video with SR = 30 %, it is evident that the color video frames recovered by QT-TNNR are visually superior to those recovered by the other methods compared, see Figs. 4–6. When the original video exhibits high missing rates (e.g., SR = 20 % and 10 %), the experimental data presented in Table 3 demonstrate that deep learning methods (3DCN and 2DCN) achieve superior recovery performance on low information datasets compared to alternative methods. This observation is further corroborated by the visual recovery outcomes in Figs. 7–8. For Mobile video with high temporal and spatial complexity, the QT-TNNR algorithm demonstrates superior restoration performance compared to alternative methods, as evidenced in Fig. 9. Additionally, it should be noted that for the large-scale dataset Bus in Table 3, our method remains the top-performing approach even under these high missing rates. Besides, for smaller datasets with substantial missing data, the results obtained by our method remain acceptable relative to T-TNN. These findings collectively reaffirm the comprehensive advantage of our method in processing large-scale datasets. Since the proposed model involves a large amount of block diagonalization during the process of doing FQT-SVD, it is not the fastest one, but it can also achieve an acceptable computational time, please refer to Table 4. Moreover, if we consider its parallel processing capability, the computational time will be significantly reduced. In the later study, we can consider the optimization algorithm mechanism to give full play to the advantages of FQT-SVD while reducing the computation.

To further examine texture detail recovery, we provide visual comparisons of key frames from the Bus (SR=50 %), News (SR=40 %), Silent (SR=30 %), Suzie (SR=20 %), Container (SR=10 %), and Mobile (SR=50 %) videos in Figs. 10 and 11. Red boxes highlight critical texture regions, while green insets display 2 × zoomed areas. Close inspection reveals that the QT-TNNR algorithm achieves a more natural restoration compared to other methods, with colors that more closely match the original images. This advantage stems from QT-TNNR's ability to preserve the intrinsic relationships among color channels through quaternion representation, combined with the truncated nuclear norm's effectiveness in maintaining numerical fidelity during reconstruction.

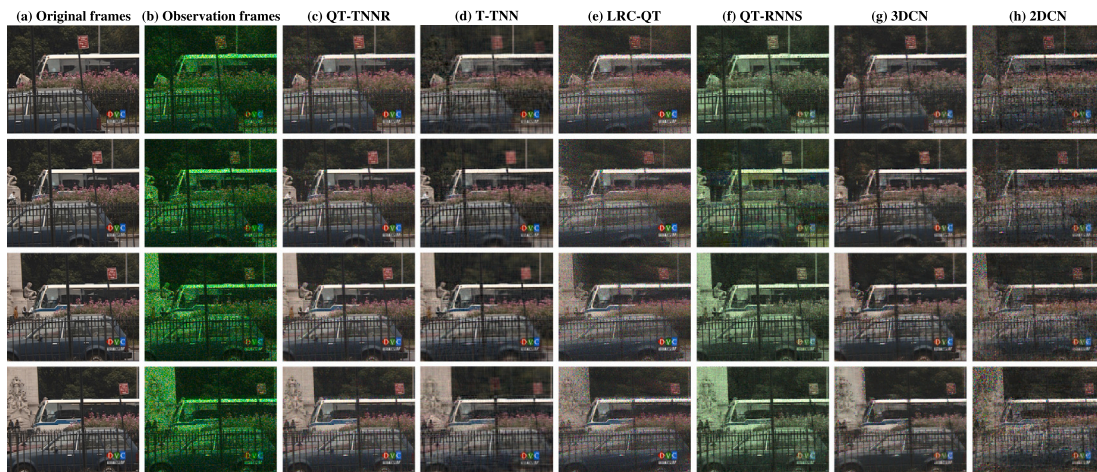


Fig. 4. The 1st, 7th, 14th and 20th frames of the Bus video are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames with SR = 50 %; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.



Fig. 5. The 1st, 7th, 14th and 20th frames of the News video are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames with SR = 40 %; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

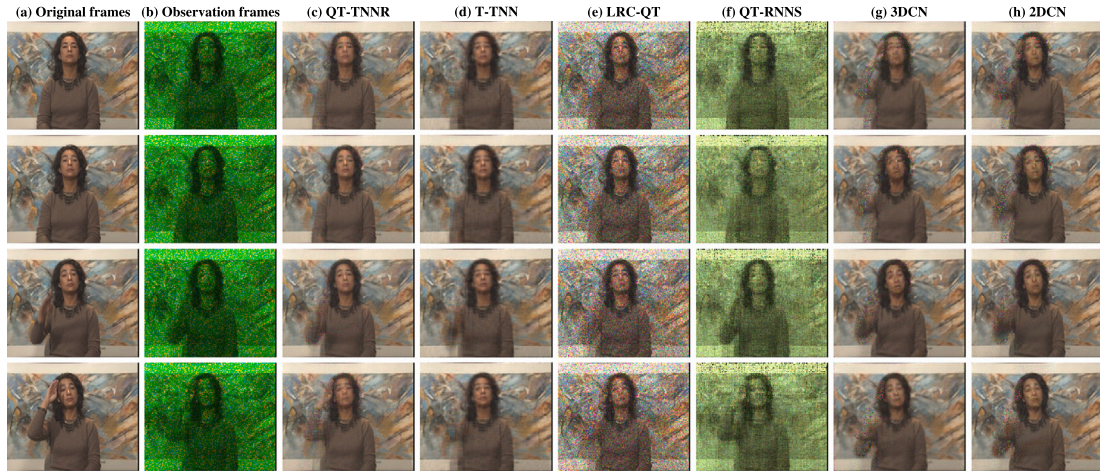


Fig. 6. The 1st, 7th, 14th and 20th frames of the Silent video are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames with SR = 30 %; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

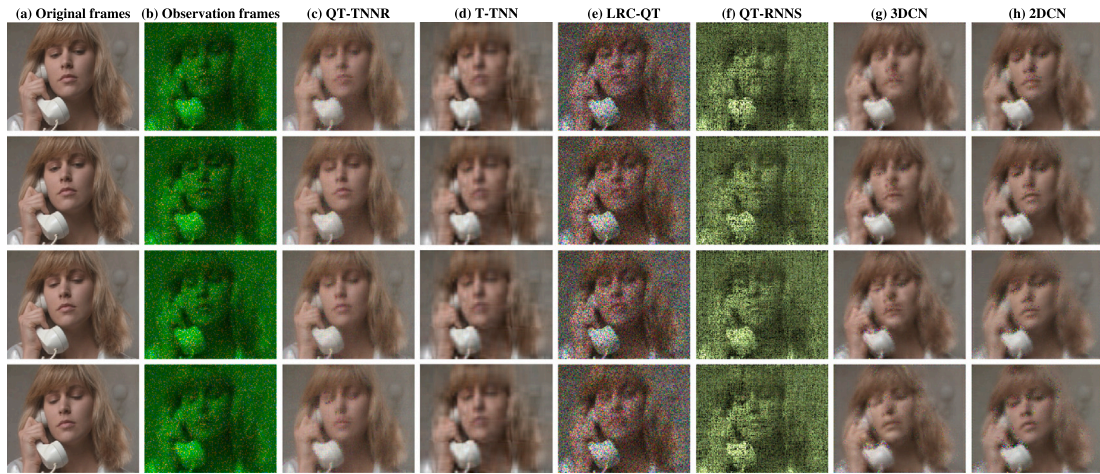


Fig. 7. The 1st, 7th, 14th and 20th frames of the Suzie video are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames with SR = 20 %; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

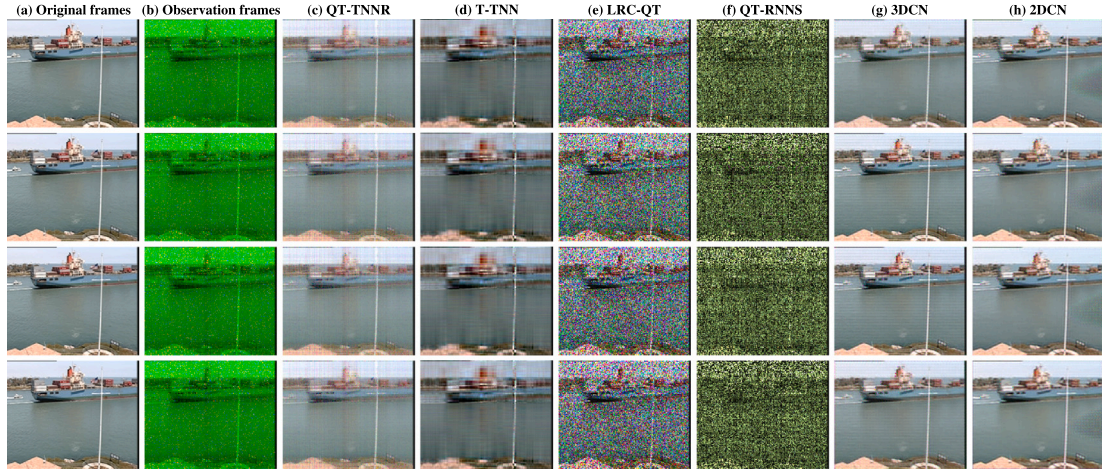


Fig. 8. The 1st, 7th, 14th and 20th frames of the Container video are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames with SR = 10 %; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

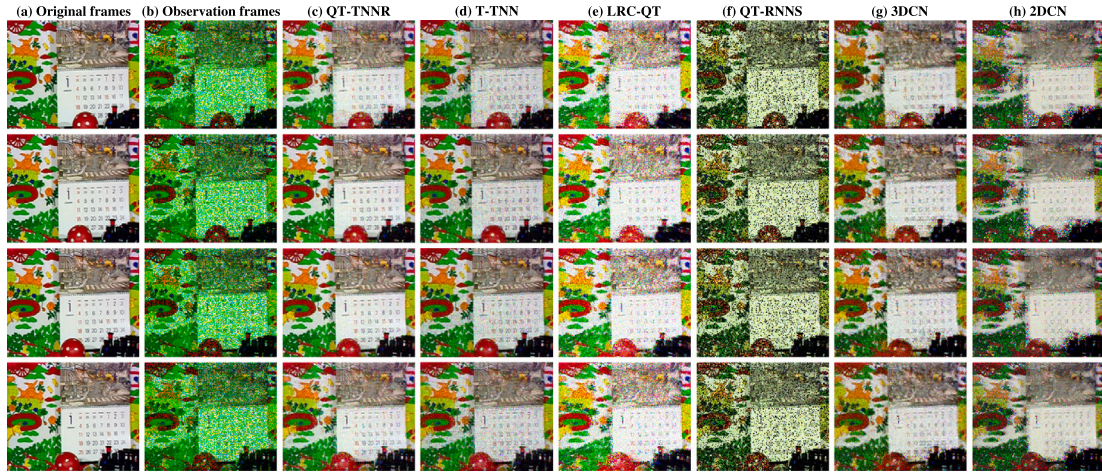


Fig. 9. The 1st, 7th, 14th and 20th frames of the Mobile video are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames with SR = 50 %; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

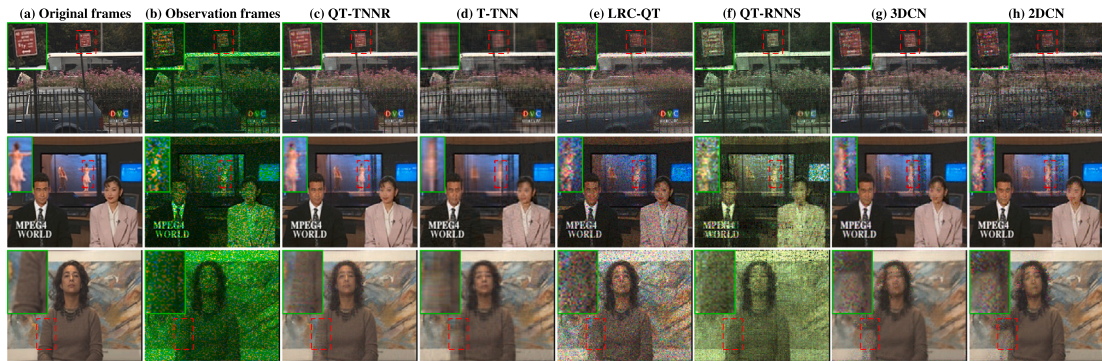


Fig. 10. The 1st frames of Bus (SR = 50 %), News (SR = 40 %), and Silent (SR = 30 %) are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

5. Conclusions

In this paper, quaternion was utilized to represent the color pixels with RGB channels, and a low-rank quaternion tensor completion method was proposed for color video recovery. The proposed method

is based on FQT-SVD and the TNNR method. Quaternion representation fully preserves the color information of the source video and handles the coupling between the color channels naturally. The quaternion tensor truncated nuclear norm was proposed to better promote the low-rankness of the tensor. To guarantee the convergence of the

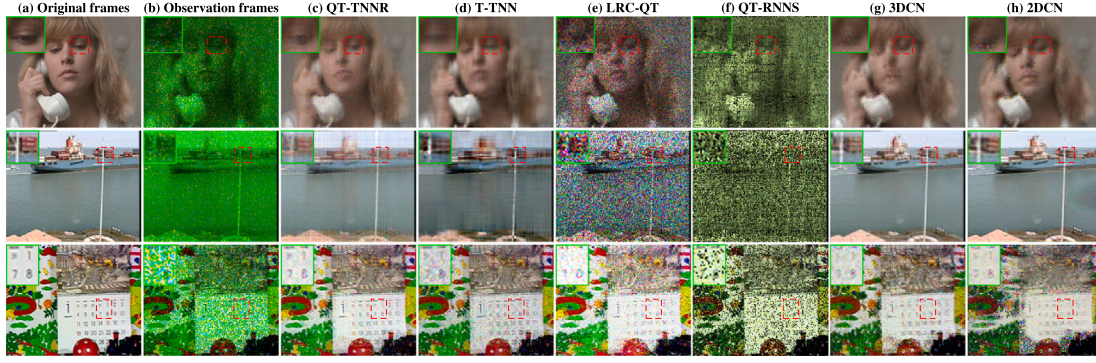


Fig. 11. The 1st frames of Suzie (SR = 20 %), Container (SR = 10 %), and Mobile (SR = 50 %) are displayed from the top to the bottom row, with each column (from left to right) representing: (a) the original frames; (b) the observation frames; (c) the recovery results of QT-TNNR; (d) T-TNN; (e) LRC-QT; (f) QT-RNNS; (g) 3DCN; (h) 2DCN.

algorithm, we adopted the ADMM framework to optimize the proposed model. Experimental results for color video recovery demonstrate the superiority of the proposed model. However, it needs to compute the FQT-SVD in each iteration, which is time-consuming for large tensors. We can consider its parallel processing capability, such that the running time will be significantly reduced.

There is still considerable potential for enhancing color video recovery through the use of quaternion tensors, which includes further investigation into more effective quaternion tensor singular value decomposition methods to boost algorithm efficiency, as well as a deeper exploration of the local properties of color video, such as sparsity, continuity, smoothness, *etc.* Moreover, the proposed completion method has broad practical application prospects. In addition to its application to the restoration of color images and videos, it also has potential applications in video surveillance, photography, and other industries.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Han Jiao: Writing – original draft, Methodology. **Qilong Liu:** Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Kun Liang:** Writing – review & editing, Visualization, Methodology. **Liyuan An:** Writing – review & editing, Formal analysis.

Acknowledgments

This research was supported by the [National Natural Science Foundation of China](#) (Grants 12461072 and 12061025), the Guizhou Provincial Basic Research Program (Natural Science) (Grant QKHJC-ZK[2023]YB245), and the Academic Young Scholars Fund of Guizhou Normal University (Grants QXXM-2021-B06 and QXXM-2021-A06).

Data availability

Data will be made available on request.

Appendix A

In this section, we prove the following equation:

$$\text{tr}(\mathcal{A} \star \mathcal{B} \star \mathcal{C}) = \text{tr}(\tilde{\mathcal{A}}^{(1)} \tilde{\mathcal{B}}^{(1)} \tilde{\mathcal{C}}^{(1)}). \quad (\text{A.1})$$

Based on (3), (4) and Definition 1, we have

$$\mathcal{A} \star \mathcal{C} = \text{fold}(\text{bcirc}(\mathcal{A})\text{unfold}(\mathcal{C}))$$

$$\begin{aligned} &= \text{fold} \left(\begin{pmatrix} \hat{\mathbf{A}}^{(1)} & \hat{\mathbf{A}}^{(n_3)} & \cdots & \hat{\mathbf{A}}^{(2)} \\ \hat{\mathbf{A}}^{(2)} & \hat{\mathbf{A}}^{(1)} & \cdots & \hat{\mathbf{A}}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\mathbf{A}}^{(n_3)} & \hat{\mathbf{A}}^{(n_3-1)} & \cdots & \hat{\mathbf{A}}^{(1)} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{G}}^{(1)} \\ \hat{\mathbf{G}}^{(2)} \\ \vdots \\ \hat{\mathbf{G}}^{(n_3)} \end{pmatrix} \right) \\ &= \text{fold} \left(\begin{pmatrix} \sum_{i=1}^1 \hat{\mathbf{A}}^{(2-i)} \hat{\mathbf{G}}^{(i)} + \sum_{i=2}^{n_3} \hat{\mathbf{A}}^{(n_3+2-i)} \hat{\mathbf{G}}^{(i)} \\ \sum_{i=1}^2 \hat{\mathbf{A}}^{(3-i)} \hat{\mathbf{G}}^{(i)} + \sum_{i=3}^{n_3} \hat{\mathbf{A}}^{(n_3+3-i)} \hat{\mathbf{G}}^{(i)} \\ \vdots \\ \sum_{i=1}^{n_3} \hat{\mathbf{A}}^{(n_3+1-i)} \hat{\mathbf{G}}^{(i)} \end{pmatrix} \right). \end{aligned} \quad (\text{A.2})$$

Suppose $\hat{\mathcal{F}} = \mathcal{A} \star \mathcal{C}$, such that

$$\hat{\mathcal{F}} = \text{fold} \left(\begin{pmatrix} \hat{\mathbf{F}}^{(1)} \\ \hat{\mathbf{F}}^{(2)} \\ \vdots \\ \hat{\mathbf{F}}^{(n_3)} \end{pmatrix} \right). \quad (\text{A.3})$$

Comparing (A.2) and (A.3), we have the following equality:

$$\begin{cases} \hat{\mathbf{F}}^{(1)} = \sum_{i=1}^1 \hat{\mathbf{A}}^{(2-i)} \hat{\mathbf{G}}^{(i)} + \sum_{i=2}^{n_3} \hat{\mathbf{A}}^{(n_3+2-i)} \hat{\mathbf{G}}^{(i)}, \\ \hat{\mathbf{F}}^{(2)} = \sum_{i=1}^2 \hat{\mathbf{A}}^{(3-i)} \hat{\mathbf{G}}^{(i)} + \sum_{i=3}^{n_3} \hat{\mathbf{A}}^{(n_3+3-i)} \hat{\mathbf{G}}^{(i)}, \\ \vdots \\ \hat{\mathbf{F}}^{(n_3)} = \sum_{i=1}^{n_3} \hat{\mathbf{A}}^{(n_3+1-i)} \hat{\mathbf{G}}^{(i)}. \end{cases} \quad (\text{A.4})$$

Due to $\text{tr}(\hat{\mathbf{A}} \pm \hat{\mathbf{G}}) = \text{tr}(\hat{\mathbf{A}}) \pm \text{tr}(\hat{\mathbf{G}})$, then

$$\begin{aligned} \text{tr}(\mathcal{A} \star \mathcal{C}) &= \text{tr}(\hat{\mathcal{F}}) = \sum_{i=1}^{n_3} \text{tr}(\hat{\mathbf{F}}^{(i)}) \\ &= \text{tr} \left(\sum_{i=1}^1 \hat{\mathbf{A}}^{(2-i)} \hat{\mathbf{G}}^{(i)} + \sum_{i=2}^{n_3} \hat{\mathbf{A}}^{(n_3+2-i)} \hat{\mathbf{G}}^{(i)} \right) \\ &\quad + \text{tr} \left(\sum_{i=1}^2 \hat{\mathbf{A}}^{(3-i)} \hat{\mathbf{G}}^{(i)} + \sum_{i=3}^{n_3} \hat{\mathbf{A}}^{(n_3+3-i)} \hat{\mathbf{G}}^{(i)} \right) \\ &\quad + \cdots + \text{tr} \left(\sum_{i=1}^{n_3} \hat{\mathbf{A}}^{(n_3+1-i)} \hat{\mathbf{G}}^{(i)} \right) \\ &= \text{tr} \left(\left(\sum_{i=1}^{n_3} \hat{\mathbf{A}}^{(i)} \right) (\hat{\mathbf{G}}^{(1)} + \hat{\mathbf{G}}^{(2)} + \cdots + \hat{\mathbf{G}}^{(n_3)}) \right). \end{aligned} \quad (\text{A.5})$$

Besides, suppose

$$\hat{\mathcal{C}} = \mathcal{B} \star \mathcal{C} = \text{fold}(\text{bcirc}(\mathcal{B})\text{unfold}(\mathcal{C}))$$

$$\begin{aligned}
&= \text{fold} \left(\begin{pmatrix} \dot{\mathbf{B}}^{(1)} & \dot{\mathbf{B}}^{(n_3)} & \dots & \dot{\mathbf{B}}^{(2)} \\ \dot{\mathbf{B}}^{(2)} & \dot{\mathbf{B}}^{(1)} & \dots & \dot{\mathbf{B}}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \dot{\mathbf{B}}^{(n_3)} & \dot{\mathbf{B}}^{(n_3-1)} & \dots & \dot{\mathbf{B}}^{(1)} \end{pmatrix} \begin{pmatrix} \dot{\mathbf{C}}^{(1)} \\ \dot{\mathbf{C}}^{(2)} \\ \vdots \\ \dot{\mathbf{C}}^{(n_3)} \end{pmatrix} \right) \\
&= \text{fold} \left(\begin{pmatrix} \sum_{i=1}^1 \dot{\mathbf{B}}^{(2-i)} \dot{\mathbf{C}}^{(i)} + \sum_{i=2}^{n_3} \dot{\mathbf{B}}^{(n_3+2-i)} \dot{\mathbf{C}}^{(i)} \\ \sum_{i=1}^2 \dot{\mathbf{B}}^{(3-i)} \dot{\mathbf{C}}^{(i)} + \sum_{i=3}^{n_3} \dot{\mathbf{B}}^{(n_3+3-i)} \dot{\mathbf{C}}^{(i)} \\ \vdots \\ \sum_{i=1}^{n_3} \dot{\mathbf{B}}^{(n_3+1-i)} \dot{\mathbf{C}}^{(i)} \end{pmatrix} \right), \quad (\text{A.6})
\end{aligned}$$

which means

$$\begin{cases} \dot{\mathbf{G}}^{(1)} = \sum_{i=1}^1 \dot{\mathbf{B}}^{(2-i)} \dot{\mathbf{C}}^{(i)} + \sum_{i=2}^{n_3} \dot{\mathbf{B}}^{(n_3+2-i)} \dot{\mathbf{C}}^{(i)}, \\ \dot{\mathbf{G}}^{(2)} = \sum_{i=1}^2 \dot{\mathbf{B}}^{(3-i)} \dot{\mathbf{C}}^{(i)} + \sum_{i=3}^{n_3} \dot{\mathbf{B}}^{(n_3+3-i)} \dot{\mathbf{C}}^{(i)}, \\ \vdots \\ \dot{\mathbf{G}}^{(n_3)} = \sum_{i=1}^{n_3} \dot{\mathbf{B}}^{(n_3+1-i)} \dot{\mathbf{C}}^{(i)}. \end{cases} \quad (\text{A.7})$$

Thus, we can convert (A.5) to

$$\begin{aligned}
\text{tr}(\dot{\mathcal{A}} \star \dot{\mathcal{B}} \star \dot{\mathcal{C}}) &= \text{tr} \left(\left(\sum_{i=1}^{n_3} \dot{\mathbf{A}}^{(i)} \right) (\dot{\mathbf{G}}^{(1)} + \dot{\mathbf{G}}^{(2)} + \dots + \dot{\mathbf{G}}^{(n_3)}) \right) \\
&= \text{tr} \left(\left(\sum_{i=1}^{n_3} \dot{\mathbf{A}}^{(i)} \right) \left(\sum_{i=1}^{n_3} \dot{\mathbf{B}}^{(i)} \right) \left(\sum_{i=1}^{n_3} \dot{\mathbf{C}}^{(i)} \right) \right). \quad (\text{A.8})
\end{aligned}$$

In the Fourier domain, it has been proved that $\bar{\mathbf{A}}^{(1)} = \sum_{i=1}^{n_3} \dot{\mathbf{A}}^{(i)}$, which implies

$$\text{tr}(\dot{\mathcal{A}} \star \dot{\mathcal{B}} \star \dot{\mathcal{C}}) = \text{tr}(\bar{\mathbf{A}}^{(1)} \bar{\mathbf{B}}^{(1)} \bar{\mathbf{C}}^{(1)}). \quad (\text{A.9})$$

The proof is accomplished.

Appendix B

Using the (A.8), we provide the mathematical derivation of $\text{tr}(\dot{\mathcal{A}}_\ell \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_\ell^H)$ as follows

$$\frac{\partial (\text{tr}(\dot{\mathcal{A}}_\ell \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_\ell^H))}{\partial \dot{\mathcal{H}}} = \frac{\partial \left(\text{tr} \left(\left(\sum_{i=1}^{n_3} \dot{\mathbf{A}}_\ell^{(i)} \right) \left(\sum_{i=1}^{n_3} \dot{\mathbf{H}}^{(i)} \right) \left(\sum_{i=1}^{n_3} \dot{\mathbf{B}}_\ell^{(i)} \right)^H \right) \right)}{\partial \dot{\mathcal{H}}}. \quad (\text{B.1})$$

Suppose

$$\dot{\mathcal{J}} = \frac{\partial \left(\text{tr}(\dot{\mathcal{A}}_\ell \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_\ell^H) \right)}{\partial \dot{\mathcal{H}}}. \quad (\text{B.2})$$

Next, we compute the partial derivatives with respect to $\dot{\mathbf{H}}^{(i)}$, such that

$$\begin{aligned}
\dot{\mathbf{T}}^{(i)} &= \frac{\partial \left(\text{tr}(\dot{\mathcal{A}}_\ell \star \dot{\mathcal{H}} \star \dot{\mathcal{B}}_\ell^H) \right)}{\partial \dot{\mathbf{H}}^{(i)}} \\
&= \frac{\partial \left(\text{tr} \left(\left(\sum_{i=1}^{n_3} \dot{\mathbf{A}}_\ell^{(i)} \right) \left(\sum_{i=1}^{n_3} \dot{\mathbf{H}}^{(i)} \right) \left(\sum_{i=1}^{n_3} \dot{\mathbf{B}}_\ell^{(i)} \right)^H \right) \right)}{\partial \dot{\mathbf{H}}^{(i)}} \\
&= \left(\sum_{i=1}^{n_3} \dot{\mathbf{A}}_\ell^{(i)} \right)^H \left(\sum_{i=1}^{n_3} \dot{\mathbf{B}}_\ell^{(i)} \right). \quad (\text{B.3})
\end{aligned}$$

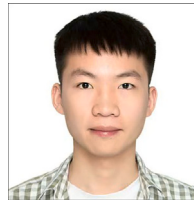
The derivative has been calculated.

References

- [1] B. Qin, M. Jin, D. Hao, Y. Lv, Q. Liu, Y. Zhu, S. Ding, J. Zhao, B. Fei, Accurate vessel extraction via tensor completion of background layer in x-ray coronary angiograms, *Pattern Recognit.* 87 (2019) 38–54.
- [2] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, J. Huang, J. Chanussot, G. Vivone, LRTCFpan: low-rank tensor completion based framework for pansharpening, *IEEE Trans. Image Process.* 32 (2023) 1640–1655.
- [3] S. Lu, P. Wang, W. Zhu, C. Dai, Y. Zhang, C. Liu, S. Dai, A nonlocal feature self-similarity based tensor completion method for video recovery, *Neurocomputing* 580 (2024) 127513.
- [4] W. Qin, H. Wang, F. Zhang, J. Wang, X. Luo, T. Huang, Low-rank high-order tensor completion with applications in visual data, *IEEE Trans. Image Process.* 31 (2022) 2433–2448.
- [5] C.-H. Lin, Y. Liu, C.-Y. Chi, C.-C. Hsu, H. Ren, T.Q.S. Quek, Hyperspectral tensor completion using low-rank modeling and convex functional analysis, *IEEE Trans. Neural Networks Learn. Syst.* (2023).
- [6] X.-Z. Li, T.-X. Jiang, L. Yang, G. Liu, Learning a more compact representation for low-rank tensor completion, *Neurocomputing* 617 (2025) 129036.
- [7] M. Wang, Z. Chen, S. Zhang, A robust low-rank tensor completion model with sparse noise for higher-order data recovery, *IET Image Proc.* 18 (12) (2024) 3430–3446.
- [8] X. Chen, K. Wang, F. Zhao, F. Deng, Q. Ye, Composite nonconvex low-rank tensor completion with joint structural regression for traffic sensor networks data recovery, *IEEE Trans. Comput. Soc. Syst.* (2024).
- [9] X. Zhang, D. Wang, Z. Zhou, Y. Ma, Robust low-rank tensor recovery with rectification and alignment, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (1) (2019) 238–255.
- [10] F. Zhang, J. Wang, W. Wang, C. Xu, Low-tubal-rank plus sparse tensor recovery with prior subspace information, *IEEE Trans. Pattern Anal. Mach. Intell.* 43 (10) (2020) 3492–3507.
- [11] C. Zou, K.I. Kou, Y. Wang, Quaternion collaborative and sparse representation with application to color face recognition, *IEEE Trans. Image Process.* 25 (7) (2016) 3287–3302.
- [12] S.-C. Pei, J.-J. Ding, J.-H. Chang, Efficient implementation of quaternion Fourier transform, convolution, and correlation by 2-d complex FFT, *IEEE Trans. Signal Process.* 49 (11) (2001) 2783–2797.
- [13] S.J. Sangwine, Fourier transforms of colour images using quaternion or hypercomplex numbers, *Electron. Lett.* 32 (21) (1996) 1979–1980.
- [14] Z. Jia, M.K. Ng, W. Wang, Color image restoration by saturation-value total variation, *SIAM J. Imag. Sci.* 12 (2) (2019) 972–1000.
- [15] Z. Jia, Q. Jin, M.K. Ng, X.-L. Zhao, Non-local robust quaternion matrix completion for large-scale color image and video inpainting, *IEEE Trans. Image Process.* 31 (2022) 3868–3883.
- [16] J. Miao, K.I. Kou, W. Liu, Low-rank quaternion tensor completion for recovering color videos and images, *Pattern Recognit.* 107 (2020) 107505.
- [17] C. Li, Y. Liu, F. Wu, M. Che, Randomized block Krylov subspace algorithms for low-rank quaternion matrix approximations, *Numer. Algorithms.* 96 (2) (2024) 687–717.
- [18] Y. Liu, F. Wu, M. Che, C. Li, Fixed-precision randomized quaternion singular value decomposition algorithm for low-rank quaternion matrix approximations, *Neurocomputing* 580 (2024) 127490.
- [19] J. Shi, X. Zheng, J. Wu, B. Gong, Q. Zhang, S. Ying, Quaternion grassmann average network for learning representation of histopathological image, *Pattern Recognit.* 89 (2019) 67–76.
- [20] F. Wu, C. Li, Y. Li, N. Tang, Complex representation matrix of third-order quaternion tensors with application to video inpainting, *J. Optim. Theory Appl.* 205 (2) (2025) 1–33.
- [21] Z. Zhou, Y. Chen, Y. Zhou, Simultaneously learning deep quaternion reconstruction and noise convolutional dictionary for color image denoising, *IEEE Trans. Emerg. Top. Comput. Intell.* (2024).
- [22] F. Wu, Y. Liu, C. Li, Quaternion tensor tri-factorization for the low-rank approximation with application to video inpainting, *Comput. Appl. Math.* 44 (2) (2025) 1–21.
- [23] J. Miao, K.I. Kou, Y. Yang, L. Yang, J. Han, Quaternion matrix completion using untrained quaternion convolutional neural network for color image inpainting, *Signal Process.* 221 (2024) 109504.
- [24] P. Wu, K.I. Kou, J. Miao, Efficient low-rank quaternion matrix completion under the learnable transforms for color image recovery, *Appl. Math. Lett.* 148 (2024) 108880.
- [25] L. Yang, K.I. Kou, J. Miao, Y. Liu, P.M. Hoi, Quaternion tensor completion with sparseness for color video recovery, *Appl. Soft Comput.* 154 (2024) 111322.
- [26] Z. Zhang, S. Aeron, Exact tensor completion using t-SVD, *IEEE Trans. Signal Process.* 65 (6) (2016) 1511–1526.
- [27] J.M. Landsberg, *Tensors: Geometry and Applications: Geometry and Applications*, vol. 128, J. Am. Math. Soc., 2011.
- [28] C.J. Hillar, L.-H. Lim, Most tensor problems are NP-hard, *J. ACM* 60 (6) (2013) 1–39.
- [29] L.R. Tucker, Some mathematical notes on three-mode factor analysis, *Psychometrika* 31 (3) (1966) 279–311.
- [30] T.G. Kolda, B.W. Bader, Tensor decompositions and applications, *SIAM Rev.* 51 (3) (2009) 455–500.
- [31] H.A.L. Kiers, Towards a standardized notation and terminology in multiway analysis, *J. Chemom.* 14 (3) (2000) 105–122.
- [32] M.E. Kilmer, C.D. Martin, Factorization strategies for third-order tensors, *Linear Algebra Appl.* 435 (3) (2011) 641–658.
- [33] M.E. Kilmer, K. Braman, N. Hao, R.C. Hoover, Third-order tensors as operators on matrices: a theoretical and computational framework with applications in imaging, *SIAM J. Matrix Anal. Appl.* 34 (1) (2013) 148–172.

- [34] Z. Zhang, G. Ely, S. Aeron, N. Hao, M. Kilmer, Novel methods for multilinear data completion and de-noising based on tensor-SVD, in: CVPR 2014, 2014, pp. 3842–3849.
- [35] J. Lou, Y.-M. Cheung, Robust low-rank tensor minimization via a new tensor spectral k -support norm, IEEE Trans. Image Process. 29 (2019) 2314–2327.
- [36] Y. Hu, D. Zhang, J. Ye, X. Li, X. He, Fast and accurate matrix completion via truncated nuclear norm regularization, IEEE Trans. Pattern Anal. Mach. Intell. 35 (9) (2012) 2117–2130.
- [37] Q. Liu, Z. Lai, Z. Zhou, F. Kuang, Z. Jin, A truncated nuclear norm regularization method based on weighted residual error for matrix completion, IEEE Trans. Image Process. 25 (1) (2015) 316–330.
- [38] C. Lee, E.Y. Lam, Computationally efficient truncated nuclear norm minimization for high dynamic range imaging, IEEE Trans. Image Process. 25 (9) (2016) 4145–4157.
- [39] S. Xue, W. Qiu, F. Liu, X. Jin, Low-rank tensor completion by truncated nuclear norm regularization, in: ICPR 2018, IEEE, 2018, pp. 2600–2605.
- [40] Z. Jia, J. Zhu, A new low-rank learning robust quaternion tensor completion method for color video inpainting problem and fast algorithms, arXiv preprint arXiv:2306.09652 (2023).
- [41] J. Miao, K.I. Kou, L. Yang, D. Cheng, Quaternion tensor train rank minimization with sparse regularization in a transformed domain for quaternion tensor completion, Knowl. Based Syst. 284 (2024) 111222.
- [42] J.A. Bengua, H.N. Phien, H.D. Tuan, M.N. Do, Efficient tensor completion for color image and video recovery: low-rank tensor train, IEEE Trans. Image Process. 26 (5) (2017) 2466–2479.
- [43] J. Pan, M.K. Ng, Block-diagonalization of quaternion circulant matrices with applications, SIAM J. Matrix Anal. Appl. 45 (3) (2024) 1429–1454.
- [44] Z. Lin, R. Liu, Z. Su, Linearized alternating direction method with adaptive penalty for low-rank representation, Adv. Neural Inf. Process. Syst. 24 (2011).
- [45] S. Boyd, N. Parikh, E. Chu, B. Peleato, J. Eckstein, et al., Distributed optimization and statistical learning via the alternating direction method of multipliers, Found. Trends Mach. Learn. 3 (1) (2011) 1–122.
- [46] J. Miao, K.I. Kou, Quaternion-based bilinear factor matrix norm minimization for color image inpainting, IEEE Trans. Signal Process. 68 (2020) 5617–5631.
- [47] J.-F. Cai, E.J. Candès, Z. Shen, A singular value thresholding algorithm for matrix completion, SIAM J. Optim. 20 (4) (2010) 1956–1982.
- [48] C. Wang, H. Huang, X. Han, J. Wang, Video inpainting by jointly learning temporal structure and spatial details, in: AAAI Conf. Artif. Intell., vol. 33, 2019, pp. 5232–5239.
- [49] S. Iizuka, E. Simo-Serra, H. Ishikawa, Globally and locally consistent image completion, ACM Trans. Graphics 36 (4) (2017) 1–14.
- [50] T. Installations, L. Line, ITU-TP 910, Subjective video quality assessment methods for multimedia applications, Recommendation ITU-T 910, 2023.

Author biography



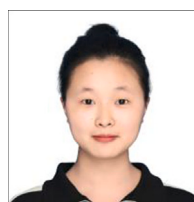
Han Jiao is currently a master's student at Guizhou Normal University. His research interests include quaternion algebra and low-rank tensor completion.



Qilong Liu received the B.S. degree in mathematics from ChuXiong Normal University, Chuxiong, China in 2009, the M.S. and Ph.D. degrees in computational mathematics from School of Mathematics and Statistics at Yunnan University, Kunming, China in 2012 and 2017, respectively. He is currently an Associate Professor with the School of Mathematical Sciences, Guizhou Normal University, Guiyang, China. His research interests include numerical algebra, tensor decomposition and image processing.



Kun Liang is currently studying for a master's degree at Guizhou Normal University. His research interests include the application of non-negative matrix factorization and non-negative tensor low-rank factorization in clustering.



Liyuan An is currently a master's student at Guizhou Normal University, with primary research interests in stochastic algorithms for solving matrix and tensor systems.